CHEMISTRY GRADE 10: INTRODUCTION TO SALTS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 2.0: Physical Chemistry
Sub-Strand	Sub-Strand 2.2: Introduction to Salts
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Types of salts (normal, acidic, basic and double salts) – Solubility of salts (chlorides, carbonates, nitrates and sulphates) – Methods of preparation of soluble salts (direct synthesis; reactions between acids and metals; acids and bases; acids and carbonates/hydrogen carbonates) – Methods of preparation of insoluble salts (direct synthesis; double decomposition/precipitation reaction) – Writing balanced chemical equations and ionic equations for salt preparation reactions – Behaviour of salts when exposed to air (hygroscopic, deliquescent and efflorescent salts) – Uses of salts (agriculture, food industry, medicine, paper industry, paints industry, glass industry, laundry and others)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) classify different salts based on their properties, b) prepare salts using appropriate methods in the laboratory, c) describe the behaviour of salts when exposed to air, d) outline applications of salts in day-to-day life, e) appreciate the applications of salts in day-to-day life.
Core Competencies	– Communication and Collaboration: The learner develops teamwork by contributing to group decision making while working with peers as they carry out laboratory experiments on salt solubility, preparation and behaviour in air, and as they brainstorm on the meaning and classification of salts using samples such as inorganic fertilisers, table salt, and commercial salts found at home and in the school laboratory. – Digital Literacy: The learner interacts with digital technology as they watch a video on the preparation of zinc sulphate ($ZnSO_4$), search for information on the applications of salts across industries (agriculture, medicine, food, glass, paper, laundry and paints), and use digital tools to record and present experimental findings.
Core Values	– Responsibility: The learner develops accountability as they take care of apparatus while using them during experiments on salt preparation and solubility, and as they properly dispose of waste after experiments, ensuring a safe and orderly laboratory environment. – Respect: The learner demonstrates respect for peers' contributions during group brainstorming activities and collaborative discussions on the classification and uses of salts. – Unity: The learner appreciates the importance of working cooperatively in laboratory groups, recognising that shared effort

	leads to more accurate and reliable experimental results.
Science & Engineering Practices	– Planning and Carrying Out Investigations: Learners design and conduct experiments to test the solubility of chlorides, carbonates, nitrates and sulphates in water, and to prepare both soluble and insoluble salts using multiple methods, recording observations systematically. – Analysing and Interpreting Data: Learners analyse solubility test results to classify salts as soluble or insoluble, and interpret mass-change data from hygroscopic, deliquescent and efflorescent salt experiments to explain behaviour in air. – Constructing Explanations: Learners use experimental evidence and balanced chemical equations (including ionic equations for precipitation reactions) to explain how different preparation methods produce salts from acids reacting with metals, bases, metal oxides, carbonates and hydrogen carbonates. – Obtaining, Evaluating and Communicating Information: Learners search for, evaluate and present information on real-world applications of salts in Kenya's agriculture (fertilisers in the Rift Valley), food industry (preservation of nyama choma and fish from Lake Victoria), medicine and manufacturing industries.
Pertinent & Contemporary Issues (PCIs)	– Environmental Issues — Safety in the Laboratory: The learner observes safety when handling acids and bases during salt preparation experiments, including correct use of protective equipment, careful handling of corrosive substances, and proper disposal of chemical waste to protect the environment. – Life Skills — Learner Rights and Responsibilities: The learner brainstorms on rights and responsibilities related to a safe learning environment when preparing salts, fostering a culture of mutual care and accountability in the laboratory. – Social Cohesion: Group-based laboratory and brainstorming activities promote inclusive participation, with learners from diverse backgrounds (urban Nairobi schools and rural Kericho or Kisumu contexts) contributing equally to shared investigations.
Career Connections	– Agricultural Scientist / Agronomist: Understanding salt chemistry underpins knowledge of inorganic fertilisers (e.g., ammonium nitrate, calcium phosphate) widely used on farms across the Rift Valley and Western Kenya to improve soil fertility and crop yields. – Pharmacist / Pharmaceutical Chemist: Many medicines used in Kenyan hospitals and clinics are salts (e.g., oral rehydration salts, iron(II) sulphate for anaemia treatment); this sub-strand builds foundational knowledge for pharmaceutical manufacturing and dispensing. – Food Scientist / Nutritionist: Salt (sodium chloride) and related compounds are used in food preservation and processing — directly relevant to Kenya's fish-processing industries along Lake Victoria and the coastal fish trade in Mombasa. – Industrial / Chemical Engineer: Salts are raw materials in Kenya's glass, paper, paint and soap industries; engineers rely on salt chemistry for quality control and process design. – Environmental Health Officer: Knowledge of salt solubility and chemical behaviour informs safe disposal of industrial chemical waste and management of saline soils in arid and semi-arid regions of Kenya.
Focus for Lessons	This unit introduces Grade 10 learners to the chemistry of salts within the broader context of Physical Chemistry, building directly on their prior learning about acids and bases. Learners investigate salt classification, solubility, preparation methods and atmospheric behaviour through a sequence of hands-on experiments, a video demonstration and research activities, all anchored in Kenyan everyday contexts such as table salt, agricultural fertilisers and food preservation. The unit emphasises scientific inquiry, collaborative laboratory work, accurate equation writing and the real-world relevance of salt chemistry to Kenya's agriculture, food, medicine and manufacturing sectors.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same — and what does understanding salts tell us about how to prepare and use them safely in Kenya?

	KICD KEY INQUIRY QUESTIONS: 1. How do salts behave when exposed to air? 2. How can salts be prepared in the laboratory?
Anchoring Phenomenon	A bag of fertiliser (calcium ammonium nitrate — CAN — commonly used on maize farms in the Rift Valley) is left open overnight in the school storeroom. The next morning, learners observe that the granules have clumped together and feel wet and sticky, even though no water was visibly added. Meanwhile, a container of washing soda (sodium carbonate decahydrate, $Na_2CO_3 \cdot 10H_2O$) left on the laboratory bench has developed a white powdery crust on its surface, and the crystals appear to have shrunk. At the same time, ordinary table salt (NaCl) in the kitchen store appears completely unchanged. This is puzzling: all three substances are salts — so why do they behave so differently when exposed to the same air? This anchoring phenomenon drives the unit investigation: learners must first understand what salts are, how they are classified, how they are prepared, and what properties they carry — before they can fully explain the contrasting atmospheric behaviours they observed.
Storyline Thread	Lesson 1 — What Is a Salt? Classification and Types: Anchored by the opening phenomenon (fertiliser, washing soda and table salt behaving differently overnight), learners brainstorm the meaning of "salt" using physical samples (table salt, CAN fertiliser, copper sulphate, sodium carbonate). They discuss and categorise salts as normal, acidic, basic and double salts, building an initial class model and posting first questions on the Driving Question Board (DQB). Lesson 2 — Solubility of Salts (Experiment 1): Learners carry out a structured solubility experiment, testing chlorides, carbonates, nitrates and sulphates in water and classifying results as soluble or insoluble. They record data in solubility tables, begin constructing a reference chart and revisit the DQB to answer: "Why does table salt dissolve in githeri water but calcium carbonate does not?" Lesson 3 — Preparing Soluble Salts I — Acid + Metal and Acid + Metal Oxide (Experiment 2a): Learners prepare zinc sulphate by reacting zinc with dilute sulphuric acid (acid + metal), collecting and testing hydrogen gas produced. They then prepare copper sulphate using copper oxide and dilute sulphuric acid (acid + metal oxide). Learners write balanced equations and connect observations to the $ZnSO_4$ video demonstration. The DQB is updated with new mechanistic understanding. Lesson 4 — Preparing Soluble Salts II — Acid + Base and Acid + Carbonate/Hydrogen Carbonate (Experiment 2b): Learners prepare sodium chloride via neutralisation (HCl + NaOH) and calcium chloride via acid + carbonate reaction (HCl + $CaCO_3$), collecting and testing CO_2 gas. They connect the CO_2 observation back to the mandazi supporting phenomenon and write full balanced equations. Learners compare all four soluble salt preparation methods using a summary flow chart. Lesson 5 — Preparing Insoluble Salts — Direct Synthesis and Double Decomposition / Precipitation (Experiment 2c): Learners prepare an insoluble salt (barium sulphate or lead(II) iodide) by mixing two soluble salt solutions (double decomposition), observing an immediate precipitate. They write ionic equations for the precipitation reaction and compare this method to direct synthesis. The DQB is updated: "How do we choose the right preparation method for a given salt?" Lesson 6 — Behaviour of Salts in Air (Experiment 3): Learners investigate hygroscopic, deliquescent and efflorescent salts by exposing anhydrous calcium chloride, sodium carbonate decahydrate and sodium chloride to air over a measured period, recording mass changes and visual observations. They match experimental findings back to the anchoring phenomenon (fertiliser, washing soda, table salt) and fully explain it using evidence. The DQB driving question receives its first complete answer. Lesson 7 — Applications of Salts in Daily Life in Kenya (Research and Reading Activity): Learners research and present on the

	uses of salts across agriculture (fertilisers in the Rift Valley), food industry (fish preservation along Lake Victoria), medicine (oral rehydration salts, iron supplements), paper, paint, glass and laundry industries. They read a structured text on salt applications and connect each use to a specific salt property learned in earlier lessons. The DQB is updated with career and societal connections. Lesson 8 — Unit Synthesis, Model Revision and Assessment: Learners revise their cumulative explanatory models (begun in Lesson 1) to produce a final annotated diagram showing salt classification, preparation pathways, atmospheric behaviour and applications — all linked back to the anchoring phenomenon. The class revisits the full DQB, confirms answered questions, flags remaining curiosities, and completes a summative assessment task requiring them to identify an unknown salt, propose its preparation method, predict its behaviour in air and describe one Kenyan industrial use.
Supporting Phenomena	– Supporting Phenomenon 1 — Solubility Surprise: A learner preparing githeri notices that table salt ($NaCl$) dissolves quickly in water when cooking, but the calcium carbonate ($CaCO_3$) used to soften maize (the traditional Kenyan "togwa" process) does not dissolve — it remains as a white suspension. Why do some salts dissolve readily while others do not? – Supporting Phenomenon 2 — Rusting Metal Roofing Sheets: Corrugated iron roofs in coastal Mombasa corrode far faster than those in Nairobi. Learners connect this to the presence of salt spray ($NaCl$ from the ocean) accelerating electrochemical reactions — illustrating that dissolved salts conduct electricity and are chemically active. – Supporting Phenomenon 3 — Zinc Sulphate Video Observation: Learners watch a video of zinc sulphate ($ZnSO_4$) being prepared by reacting zinc metal with dilute sulphuric acid. Bubbles of hydrogen gas are produced and the colourless solution that remains evaporates to form white crystals — a vivid, puzzling transformation from solid metal and liquid acid to a completely new solid salt. – Supporting Phenomenon 4 — Baking Powder in Mandazi: When a Nairobi street vendor adds baking powder (sodium hydrogen carbonate, $NaHCO_3$) to mandazi dough and fries it, the dough rises and carbon dioxide gas is released. Learners recognise this as a salt reacting under heat — connecting salt chemistry to everyday Kenyan food preparation.

LESSON 1 (80 minutes): What Is a Salt? Classification and Types	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the anchoring phenomenon and establish foundational understanding of what salts are and how they are classified into normal, acidic, basic and double salts.
Knowledge	A salt is a substance formed when a hydrogen atom in an acid is fully or partially replaced by a positive ion. Salts are classified as normal salts (all hydrogen replaced), acidic salts (hydrogen partially replaced), basic salts (contain OH^- or O^{2-} in addition to the anion), and double salts (contain two different cations or anions).
Skills	Brainstorming and group discussion to establish the meaning of salt; categorising physical samples of salts; constructing an initial class model; posting evidence-based questions on the Driving Question Board (DQB).
Attitudes	Curiosity about everyday phenomena involving salts; responsibility in handling laboratory samples safely; appreciation of the role of salts in Kenyan agriculture, food and industry.
Key Inquiry Question	How do salts behave when exposed to air?
Purpose in Storyline	This is the opening lesson of the unit. It launches the anchoring phenomenon — fertiliser granules clumping, washing soda crystals developing a white crust, and table salt remaining unchanged — and asks learners to confront the puzzle: all three are salts, so why do they behave so differently? By the end of this lesson learners have built a shared definition of 'salt', sorted physical samples into classification categories, constructed an initial class model, and posted their first wondering questions on the DQB. Every subsequent lesson will return to this phenomenon and this board.
Safety Notes	All physical samples (copper sulphate, sodium carbonate, CAN fertiliser, table salt) should be pre-measured into sealed, labelled specimen tubes or small zip-lock bags before the lesson. Learners must NOT taste any sample. Copper sulphate is toxic — learners must wash hands thoroughly after handling. CAN fertiliser is an oxidiser — keep away from flames. Lab coats and safety goggles are required throughout the lesson. Dispose of copper sulphate waste in the designated chemical waste bin, not the sink.

B. LESSON OVERVIEW
Learners encounter the anchoring phenomenon for the first time: a bag of CAN fertiliser left open overnight in the school storeroom has clumped and feels wet; washing soda on the lab bench has shrunk and developed a white powdery crust; table salt in the kitchen store is completely unchanged. Through structured brainstorming with four physical samples (table salt, CAN fertiliser, copper sulphate, sodium carbonate), guided discussion and a sorting activity, learners build a shared definition of 'salt' as a product of acid–base neutralisation in which hydrogen atoms are replaced by positive ions. They then categorise salts as normal, acidic, basic and double salts, construct an initial class visual model, and open the unit's Driving Question Board (DQB) by posting their first questions and partial explanations about the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are presented with three labelled photographs and/or	Display the three samples/photographs prominently	Think–Pair–Share Prediction Protocol: This strategy activates	Observable indicator: Every learner has a written prediction in

📹 **VIDEO:** [Ratios on coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	physical displays: (A) a torn-open bag of CAN fertiliser granules that appear wet and clumped; (B) a dish of washing soda ($Na_2CO_3 \cdot 10H_2O$) crystals with a white powdery crust; (C) a salt shaker of ordinary table salt, unchanged. They work in pairs for 5 minutes to write silent individual responses to the prompt: 'All three substances are called salts. Predict why the fertiliser granules clumped together overnight, the washing soda formed a white crust, but the table salt stayed the same. Use any science you already know.' Pairs then share predictions with a neighbouring pair (Think–Pair–Share). One spokesperson per group of four shares one idea with the class. Teacher records ALL predictions — without correcting — on the whiteboard under the heading 'Our Predictions (Day 1)'. Learners record their own predictions in their science notebooks.	before learners enter the room so curiosity is immediate. Say: 'Before I tell you anything, I want your brain's first guess — there are no wrong answers yet.' Distribute the individual prediction slips. Allow 3 minutes of silent writing — enforce no talking. After Think–Pair–Share, cold-call FOUR groups (not volunteers) using the class register: 'Group 3, what was your prediction?' After each response, paraphrase back: 'So you are saying that maybe the fertiliser pulled water from the air — is that right?' Apply 10–12 seconds of WAIT TIME after each cold-call before accepting answers. Record every prediction verbatim on the board. Close the phase by saying: 'Hold onto these ideas — by the end of this unit, you will be able to explain exactly why each of these three salts behaved the way it did. That is our investigation.'	prior knowledge about salts from everyday life (cooking, farming, laboratory) before instruction, reveals misconceptions the teacher can address across the unit, and gives every learner a personal stake in the phenomenon by committing their prediction to writing. Recording predictions publicly creates accountability and a reference point for revision.	their notebook before whole-class sharing begins. Teacher scans notebooks during pair-share to check engagement. Key misconception to listen for: learners conflating 'salt' solely with table salt (NaCl) — note this for Phase 3. Collect 3–4 prediction slips from different ability groups to inform pacing of Phase 3.
Observe Phase 📹 **VIDEO:** [Ratios on coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group of four learners receives a tray containing four labelled specimen tubes: (1) Table salt — NaCl; (2) CAN fertiliser granules — $Ca(NO_3)_2$ / NH_4NO_3 mixture; (3) Copper sulphate crystals — $CuSO_4 \cdot 5H_2O$; (4) Washing soda crystals — $Na_2CO_3 \cdot 10H_2O$. Learners observe each sample using sight and touch ONLY (no tasting) and complete a structured observation table in their notebooks with columns: Sample name | Colour | Texture | Crystal shape (if visible) | Any other observations | 'Is this a salt? My evidence'. After individual observation (5 min), groups discuss and reach a consensus answer to: 'What do these four	Before distributing trays, say: 'You are about to handle real laboratory chemicals. Copper sulphate is toxic — do NOT taste anything. Lab coats and goggles on now.' Circulate every 90 seconds, crouching to group level. Prompt deeper observation with: 'What do you notice about the crystal shape of the copper sulphate compared to the table salt?' and 'If these are all salts, what acid do you think each one came from?' After groups record consensus statements, cold-call THREE groups (rotate from the register): 'Group 5, read me your consensus statement exactly as you wrote it.' Apply 12 seconds WAIT TIME. After each response, write key	Structured Observation Table with Consensus Protocol: Requiring learners to observe before receiving the definition ensures they build their own inductive understanding ('these things share properties') before deductive instruction. The consensus statement forces groups to negotiate language, which mirrors scientific practice. Connecting each sample back to an acid and a positive ion previews the classification work in Phase 3 and ties the samples to the phenomenon.	Observable indicator: Completed observation tables with all four columns filled for each sample, plus a consensus statement. Teacher checks that learners are connecting observations to the definition (e.g. 'NaCl comes from HCl — H replaced by Na^+'). Listen for groups who cannot identify the acid source — these groups will need targeted support in Phase 3. Indicator of readiness to move: at least 3 of 4 groups have a consensus statement that references ion replacement.

	substances have in common that makes them all called salts?' Groups record their consensus statement. Teacher then shares the KICD-mandated information card: 'A salt is a substance that is formed when a hydrogen atom in an acid is fully or partially replaced by a positive ion.' Learners annotate their observation tables with this definition and identify which acid and which positive ion they think might have formed each sample — making a first attempt even if uncertain.	phrases on the board. Do NOT confirm or deny yet — say: 'Interesting — let's test that idea.' Transition by holding up the KICD definition card: 'Here is how Kenyan chemists define a salt. Take 90 seconds to read it silently, then annotate your table.'		
Explain Phase 📹 **VIDEO:** Ratios on coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Teacher-facilitated whole-class discussion builds from the observation consensus statements toward the formal classification of salts. Learners receive or draw a Classification Anchor Chart template in their notebooks with four columns: Normal Salt | Acidic Salt | Basic Salt | Double Salt. Teacher provides the KICD definitions and worked examples for each type, connecting each to the four physical samples and to the anchoring phenomenon. Learners then work in groups to complete a sorting card activity: 16 salt formula cards (e.g. NaCl, $NaHCO_3$, $Pb(OH)Cl$, $KAl(SO_4)_2{\cdot}12H_2O$, $KHSO_4$, $CuSO_4$, NH_4NO_3, Na_2SO_4, $Ca(H_2PO_4)_2$, $Mg(OH)Cl$, $Fe_2(SO_4)_3$, $NaHSO_4$, $CaCO_3$, $ZnCO_3$, $KCr(SO_4)_2{\cdot}12H_2O$, Na_2CO_3) are sorted into the four categories with written justification for each. Groups then connect the activity back to the phenomenon: 'Which category does CAN fertiliser (NH_4NO_3) belong to? What about washing soda (Na_2CO_3)? What about table salt (NaCl)?' Each group writes a 2-	Open Phase 3 by pointing to the predictions board: 'Let's revisit your predictions. Now that you have a definition and have observed the samples, which predictions do you want to revise?' Cold-call TWO learners (choose learners who had very different predictions): 'Amina, you predicted the fertiliser absorbed water from the soil — would you revise that now, or keep it?' WAIT 12 seconds. Deliver the classification content using the four physical samples as concrete referents throughout: 'When HCl reacts with NaOH, ALL the hydrogen is replaced — NaCl is a normal salt.' Write each definition with an example equation on the board. After card sort, cold-call FIVE different groups for five different cards — include at least one acidic and one basic salt. When a group mis-sorts, say: 'That is a really common placement — can anyone tell us what additional information we need to check before we decide?' WAIT 10 seconds. Close by returning to the phenomenon: 'So we now know NaCl is a normal salt, Na_2CO_3 is a normal salt, and	Classification Anchor Chart + Card Sort: The anchor chart creates a permanent visual reference that learners will revisit and annotate across all 30 lessons. The card sort makes classification an active, error-tolerant process rather than passive note-taking, and the requirement for written justification develops scientific argumentation. Returning the card sort results to the phenomenon ('all normal salts — but different behaviours') creates productive confusion that motivates the rest of the unit.	Observable indicator: Each group's card sort is fully completed with written justification for every card. Teacher photographs each group's sorted cards before cards are returned. Exit check: teacher poses one oral question to the whole class using a show-of-hands agree/disagree response: 'Sodium hydrogen carbonate, $NaHCO_3$, is a normal salt — agree or disagree?' (Correct: disagree — it is an acidic salt because one H remains.) Count the disagree hands; if fewer than 60% respond correctly, re-teach acidic salts before closing Phase 3. Learners who correctly disagree must justify orally — cold-call TWO of them.

	sentence explanation connecting category to the phenomenon. Groups share, class agrees on correct placements, teacher corrects misconceptions using the board.	NH_4NO_3 is a normal salt. They are all normal salts — yet they behaved differently overnight. That tells us classification alone does NOT explain the phenomenon. What else must we investigate?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Ratios on coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Teacher formally introduces the unit Driving Question Board (DQB) — a large sheet of manila paper or whiteboard section that will remain displayed for all 30 lessons. The unit Driving Question is read aloud together: 'Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same — and what does understanding salts tell us about how to prepare and use them safely in Kenya?' Each learner receives three sticky notes (different colours: pink = question I still have; yellow = something I observed that I cannot yet explain; green = something I now understand). Learners write independently for 4 minutes, then place their sticky notes on the DQB in designated zones. The class does a 'gallery walk' of 3 minutes reading each other's notes. Teacher reads 5–6 selected notes aloud and clusters them under emerging theme headings: 'What makes a salt absorb water?', 'How does a salt form?', 'What are salts used for in Kenya?', 'How do we prepare salts safely?'. These clusters become the sub-questions that will guide subsequent lessons. Teacher adds the lesson-specific question: 'We now know what salts are and how to classify them — but classification alone does not explain why these three salts behaved differently. What	Introduce the DQB by saying: 'This board is our class's investigation wall. Every lesson, we will add to it, revise it, and answer parts of it. By Lesson 30 it should be nearly full of answers — but today we start with mostly questions, and that is exactly right.' Model filling in one sticky note yourself ('I notice that all three salts are normal salts, but they behaved differently — I wonder if solubility matters') and place it on the board. Allow 4 minutes of silent independent sticky-note writing — no talking. During gallery walk, prompt deeper reading: 'Find one note that asks a question you also have — give that person a silent thumbs up.' After clustering, say: 'Look at these clusters — this is YOUR unit roadmap. Each cluster will become a future lesson. Today's cluster is: What is a salt and how do we classify it? — and we have partially answered it. The fertiliser cluster and the white-crust cluster are still open. That is what drives us forward.' Take a photograph of the DQB for the ARES platform record.	Driving Question Board with Colour-Coded Sticky Notes: The three-colour system (question / unexplained observation / understood concept) externalises metacognition and makes the boundary between known and unknown visible to the whole class. Clustering notes into sub-questions transforms learner curiosity into a structured investigation roadmap, giving learners agency over the unit direction. Returning to the DQB every lesson ensures the phenomenon remains the anchor, not the textbook.	Observable indicator: Every learner places at least one sticky note in each of the three colour zones. Teacher checks that pink (question) notes are phenomenon-linked — not generic ('I wonder what a salt is' is too vague; 'I wonder why CAN fertiliser clumps but table salt does not' is phenomenon-linked). Teacher uses the yellow (unexplained observation) notes as a pre-assessment of observational vocabulary and scientific thinking to inform grouping decisions for Lesson 2's solubility experiment.

	properties of salts do we still need to investigate?'			
Model Building Phase 📹 **VIDEO:** Ratios on coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner draws an Initial Class Model in their science notebook. The model must include: (1) A diagram or annotated sketch showing the three phenomenon substances (CAN fertiliser bag, washing soda dish, table salt shaker) with their chemical formulas labelled; (2) Arrows or labels showing what they currently know or hypothesise is happening to each substance in air; (3) A box labelled 'What my model can explain today' and a box labelled 'What my model CANNOT yet explain'; (4) The KICD definition of salt written at the top as the model's foundation. Learners share models in pairs and give each other one 'warm' (something the model shows well) and one 'wonder' (a question the model raises). Groups then select one model to be the 'class anchor model' — photographed and displayed alongside the DQB. Teacher leads a brief whole-class debrief: 'What does our class model show well today? What is missing? What do we need to find out in the next lesson to improve it?' Learners record one specific revision goal in their notebooks: 'In the next lesson I want to add __________ to my model.'	Say: 'In science, a model is our best current explanation — it is not a perfect picture, it is a thinking tool. Your model today will be incomplete — that is the point. We will revise it every lesson.' Circulate during individual model drawing; prompt with: 'What do your arrows mean? Can you label them with a word?' and 'Your model shows the fertiliser getting wet — but your model does not yet show WHY. Can you add a question mark there to show that is still open?' After warm-and-wonder pair feedback, cold-call THREE pairs to share one 'wonder' each. WAIT 10 seconds after each. Write the three 'wonders' on the board — check whether any match DQB sticky notes ('Look — Kariuki's wonder matches this pink note on our DQB. That tells us our model and our board are aligned.'). Close the lesson: 'Today you built your first model of this phenomenon. It is incomplete — and that is exactly where we want to be at the end of Lesson 1. Lesson 2 will give you new evidence about solubility that will force you to revise this model. Keep your notebook safe.'	Initial Anchor Model with Warm-and-Wonder Protocol: Beginning the unit with an explicit, incomplete model establishes the norm that scientific models evolve with evidence — a core NGSS Science and Engineering Practice (Developing and Using Models). The 'What my model cannot explain' box deliberately institutionalises productive uncertainty, preventing learners from treating their first attempt as final. The warm-and-wonder protocol builds the collaborative scientific discourse norms needed for the remaining 29 lessons.	Observable indicator: Every learner has a drawn model with the three phenomenon substances labelled with formulas, at least one arrow or annotation per substance, and both the 'can explain' and 'cannot explain' boxes completed. Teacher collects 6 notebooks (two from each ability band) at the end of the lesson to assess: (a) whether the KICD definition is correctly recorded; (b) whether classification vocabulary (normal, acidic, basic, double) appears in the model; (c) the quality of what is listed in the 'cannot explain' box — rich 'cannot explain' entries (e.g. 'I cannot explain why CAN absorbs water but NaCl does not') indicate readiness for Lesson 2's solubility investigation.

D. TEACHER REFLECTION

After delivering this lesson, reflect on the following questions in your professional journal:

1. PHENOMENON LAUNCH: Did the three-sample display generate genuine curiosity, or did learners treat it as a routine lab activity? If engagement was low, consider bringing a physical torn CAN fertiliser bag and actual clumped granules from a Rift Valley farmer next time.

2. PRIOR KNOWLEDGE: What did learners' prediction slips reveal about their existing understanding of salts? Did many learners equate 'salt' only with table salt (NaCl)? If so, how will you address this salt-as-food misconception explicitly in Lesson 2?

3. CARD SORT ACCURACY: Which of the four salt types — normal, acidic, basic, double — caused the most mis-sorting? If acidic salts (e.g. $NaHCO_3$, $KHSO_4$) were widely mis-sorted, plan a 10-minute retrieval activity at the start of Lesson 2 using two examples from Kenya's water treatment context (alum is a double salt used to purify drinking water in Nairobi's treatment plants).

4. DQB QUALITY: Were learners' pink (question) sticky notes genuinely phenomenon-linked, or were they generic textbook questions? If generic, model a higher-quality phenomenon-linked question at the start of Lesson 2 ('Why does the white crust on the washing soda feel different from the original crystal?') to raise the standard.

5. MODEL COMPLETENESS: From the 6 collected notebooks, were learners able to distinguish between what the classification tells them and what remains unexplained? If learners wrote 'I cannot explain anything' in the 'cannot explain' box, they may need more scaffolding in how to articulate productive uncertainty — use a sentence stem in Lesson 2: 'My model can explain ___________ but cannot yet explain why ___________.'

6. TIMING: This lesson is designed for 80 minutes. If the card sort ran over time, consider splitting the card sort into a homework task (12 cards at home, 4 in class) for a 40-minute lesson context. Note which phase consumed the most time and plan accordingly for Lesson 2.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed four physical salt samples (table salt, CAN fertiliser granules, copper sulphate crystals, washing soda crystals) and recorded differences in colour, texture, crystal shape and appearance. They also observed the anchoring phenomenon — the clumped fertiliser, white-crusted washing soda and unchanged table salt — and noted that all three are salts despite behaving differently overnight.
What did I learn?	A salt is a substance formed when a hydrogen atom in an acid is fully or partially replaced by a positive ion. Salts are classified as: normal salts (all hydrogen atoms replaced, e.g. $NaCl$, Na_2SO_4), acidic salts (hydrogen partially replaced, e.g. $NaHCO_3$, $NaHSO_4$), basic salts (contain OH^- or O^{2-} in addition to the salt anion, e.g. $Pb(OH)Cl$, $Mg(OH)Cl$), and double salts (contain two different cations or two different anions, e.g. $KAl(SO_4)_2 \cdot 12H_2O$ — alum). CAN fertiliser (NH_4NO_3), washing soda (Na_2CO_3) and table salt ($NaCl$) are all normal salts — yet they behave differently in air, which means classification alone does not explain the phenomenon.
How does this explain the phenomenon?	Learners cannot yet fully explain WHY the three salts behaved differently overnight. They have established WHAT a salt is and HOW salts are classified, but the atmospheric behaviour (hygroscopic, deliquescent, efflorescent properties) requires understanding of solubility, water of crystallisation and vapour pressure — concepts that will be built in subsequent lessons. The DQB captures these open questions as the engine of the unit investigation.

LESSON 2 (40 minutes): Solubility of Salts — Experiment 1	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners carry out a structured solubility experiment, classify salts as soluble or insoluble, and use their findings to begin explaining why table salt dissolves in cooking water while other salts do not — advancing the unit investigation into the contrasting behaviour of fertiliser, washing soda and table salt.
Knowledge	Learners will be able to state the general solubility rules for chlorides, carbonates, nitrates and sulphates in water, and distinguish between soluble and insoluble salts using experimental evidence.
Skills	Learners will measure and record mass changes, observe dissolving behaviour, construct a solubility reference chart, and communicate findings in a structured data table.
Attitudes	Learners will demonstrate careful and accurate observation, honest recording of unexpected results, and willingness to revise their predictions when evidence contradicts prior ideas.
Key Inquiry Question	How do salts behave when exposed to air? (This lesson addresses the foundational solubility behaviour that underpins later atmospheric behaviour investigations.)
Purpose in Storyline	Lesson 1 established that NaCl, CAN and washing soda are all normal salts, yet behave differently. Before learners can explain the overnight phenomenon, they need to understand how salts interact with water — starting with liquid water in this lesson, before moving to water vapour in Lesson 6. This lesson provides the first row of evidence: solubility data.
Safety Notes	Avoid ingesting any chemicals. Copper sulphate is toxic — learners must wear gloves when handling it. Barium chloride (if used) is toxic; substitute with sodium chloride if unavailable. Wash hands thoroughly after the experiment. Dispose of solutions into the designated waste container, not the sink. Supervise learners when handling glassware.

B. LESSON OVERVIEW
Learners begin by predicting which salts will dissolve in water based on prior knowledge and the phenomenon context. They then carry out a structured experiment, adding measured amounts of seven salts — sodium chloride, potassium nitrate, calcium carbonate, copper sulphate, lead sulphate, sodium carbonate and ammonium nitrate (representing CAN) — to water in test tubes, observing and recording results. In the Explain phase, they construct a group solubility reference chart and derive general solubility rules. They connect findings to the DQB question about why githeri salt dissolves but limestone does not, and update their cumulative models with a solubility column.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** More functional groups	Learners are shown seven labelled sample vials containing: sodium chloride ($NaCl$), potassium nitrate (KNO_3), calcium carbonate	Display the seven vials and point to the CAN fertiliser sample. Say: 'Yesterday we said CAN, washing soda and table salt are all normal	Prediction table with S/I classification and written reasoning activates prior knowledge from Lesson 1 and creates cognitive	Teacher circulates and reads prediction tables. Listen for misconceptions such as 'all white salts dissolve' or 'coloured salts do

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chargaff's Base-Pairing Rules - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	($CaCO_3$), copper sulphate ($CuSO_4$), lead sulphate ($PbSO_4$), sodium carbonate (Na_2CO_3) and ammonium nitrate (NH_4NO_3). The teacher reminds them of the phenomenon: the farmer in the Rift Valley noticed CAN fertiliser (NH_4NO_3) clumped overnight — could its ability to interact with water be linked to how easily it dissolves? Learners individually complete a Predict column in their lab tables, writing S (soluble) or I (insoluble) for each salt and one sentence of reasoning. They then share predictions with a partner, noting agreements and disagreements before the teacher takes a quick show-of-hands class poll for each salt.	salts. Today we start digging into why they behave differently — and we begin with something more basic: do they even dissolve in water? Look at these seven salts. Before you touch anything, fill in your Predict column.' Allow 3 minutes of individual thinking. WAIT TIME: after calling on a learner, pause 5 seconds before accepting the answer to encourage full sentences. After the partner discussion, ask: 'Who predicted that ammonium nitrate would dissolve? Who said it would not? Keep that in mind — your evidence from the experiment will tell you who is right.' Do NOT confirm or deny predictions yet.	dissonance when predictions are later contradicted by evidence. The phenomenon anchor (CAN fertiliser) ensures predictions are not abstract.	not dissolve.' Note which learners reason from salt type (e.g. 'it is a nitrate so it dissolves') versus appearance alone — this informs grouping for the Observe phase.
Observe Phase 📹 **VIDEO:** More functional groups Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chargaff's Base-Pairing Rules - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	In groups of four, learners conduct the structured experiment. Each group receives: seven test tubes in a rack, a 250 mL beaker of distilled water, a 5 mL measuring cylinder, a spatula, a glass stirring rod, safety gloves and a data table printed on A3 paper. Procedure: (1) Add 5 mL of distilled water to each test tube. (2) Add one level spatula tip (approximately 0.1 g) of each salt, one salt per test tube. (3) Stir each tube ten times with the glass rod, rinsing between salts. (4) Observe after stirring and again after 2 minutes of standing. (5) Record in the Observe column: colour of solution or mixture, whether the solid disappeared (dissolved), and whether the mixture remained cloudy (insoluble). Learners photograph or sketch each tube. They also note the colour of the copper sulphate solution and the colourlessness of the others — this becomes relevant when	Before learners begin, demonstrate the technique with one extra sample of NaCl: 'Watch how I add just one spatula tip — too much and everything looks insoluble because we have exceeded the solubility limit. That is a source of error we want to avoid.' Circulate during the experiment. When a group observes that $CaCO_3$ stays cloudy, crouch to their level and ask: 'What does that cloudiness tell you about whether it has dissolved?' WAIT TIME: 7 seconds. If learners say 'it did not dissolve,' ask: 'How would you write that in your table — and what does that mean for limestone in river water back home?' At the 10-minute pause, ask the whole class: 'One group — tell me your most surprising result so far.' Do not validate or correct; just say: 'Interesting — finish the experiment and see if the evidence holds.'	Structured observation table with explicit columns (prediction, observation after stirring, observation after 2 minutes, conclusion) separates the act of observing from the act of concluding. Pausing mid-experiment to share surprises models scientific discourse and keeps the phenomenon visible.	Teacher checks that learners are recording observations rather than jumping to conclusions in the Observe column. Ask targeted groups: 'What exactly do you see — describe the mixture without using the word dissolved.' Listen for use of descriptive language such as cloudy, clear, coloured, suspended. Flag groups that record only S or I in the observation column — prompt them to add a visual description.

	discussing identification later. The teacher pauses the class at the 10-minute mark and asks groups to share one surprising observation before they continue.			
Explain Phase 📹 **VIDEO:** More functional groups Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chargaff's Base-Pairing Rules - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups compile their data onto a shared class chart drawn on the board (or a projected template). Each group reports one salt result; the teacher fills in the class table. Learners then work in pairs to identify patterns: they highlight all nitrates in one colour, all chlorides in another, all carbonates in a third, and all sulphates in a fourth. From the patterns, pairs draft two to four solubility rules in their notebooks. The teacher then introduces the standard solubility rules as a structured reading card (one paragraph, Kiswahili key terms in brackets): all nitrates are soluble; most chlorides are soluble except lead(II) chloride and silver chloride; most sulphates are soluble except lead(II) sulphate and barium sulphate; most carbonates are insoluble except sodium, potassium and ammonium carbonates. Learners compare their drafted rules to the reading card, annotate differences in red pen and add any missing rules. Finally, the class builds a group solubility reference chart on A3 paper (one per group) to be displayed in the lab and used in later lessons. Learners then explicitly answer the DQB question posted in Lesson 1: 'Why does table salt dissolve in githeri water but calcium carbonate does not?' — writing a two-sentence evidence-based answer in their notebooks.	After groups share data, draw attention to the class table: 'Look at the nitrate column — what do you notice? All of them?' WAIT TIME: 5 seconds. 'Now look at the carbonate column.' After pairs share their drafted rules, say: 'Scientists spent years collecting data like yours to write the solubility rules — let us see how close you got.' Distribute the reading card. After learners compare, ask: 'Which rule surprised you most and why?' WAIT TIME: 5 seconds. When connecting to the phenomenon, say: 'A Rift Valley farmer adds CAN fertiliser to soil that is often wet. Our data says ammonium nitrate is soluble. What does that mean for how quickly the fertiliser can reach the plant roots?' WAIT TIME: 7 seconds. Accept two to three responses. Close the explain phase by saying: 'Write your two-sentence answer to the githeri question now — you will need it when we answer the big driving question in Lesson 6.'	Colour-coded pattern identification (nitrates, chlorides, sulphates, carbonates) is a disciplinary literacy move that makes the structure of solubility rules visible before they are stated. Comparing learner-generated rules to the canonical rules develops metacognitive awareness. Connecting to githeri and Rift Valley farming keeps the phenomenon central.	Collect the two-sentence DQB answers at the end of the phase. Look for: (1) use of evidence from the experiment, (2) correct use of the word soluble or insoluble, (3) reference to NaCl versus $CaCO_3$ specifically. Learners who write only 'because NaCl dissolves and $CaCO_3$ does not' without referencing the experiment are identified for targeted follow-up at the start of Lesson 3.
Driving	Learners return to the DQB	Point to the DQB and say: 'In	The DQB move makes the	Read all orange sticky notes

Question Board (DQB) Creation 📹 **VIDEO:** [More functional groups] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chargaff's Base-Pairing Rules - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	established in Lesson 1. Each group receives two sticky notes: one green and one orange. On the green note, they write one question that this lesson has answered — for example, 'Why does table salt dissolve in githeri water but calcium carbonate does not?' On the orange note, they write one new question raised by their observations — for example, 'If ammonium nitrate is so soluble, does it also dissolve in water vapour in the air, not just liquid water?' Groups stick their notes in the correct DQB columns (Answered / New Questions). The teacher reads selected orange notes aloud and maps them to future lessons: solubility in water vapour leads to Lesson 6 (behaviour in air); how salts are made leads to Lessons 3 to 5.	Lesson 1 you posted questions about why the three salts behaved differently overnight. Today you have new evidence. Which question can we now move to the Answered column?' After learners identify the githeri question, ask: 'But does solubility in liquid water explain the overnight clumping of CAN? Think carefully.' WAIT TIME: 10 seconds. When learners express uncertainty — 'there was no liquid water in the storeroom' — say: 'Excellent. That tension is exactly what we need to keep investigating. Write that as your orange question.' Read two or three orange questions aloud: 'These are the questions that will drive our next four lessons. Keep them.'	boundary between what is now known and what remains unknown visible and public. Mapping orange questions to future lessons gives learners a sense of the investigation arc and maintains the phenomenon as the unresolved centre of the unit.	before the next lesson. Questions that remain purely about definitions (e.g. 'What is a salt?') indicate learners who did not consolidate Lesson 1 understanding — plan a brief re-teach hook at the start of Lesson 3. Questions that point toward water vapour or atmospheric moisture indicate learners who are making productive cross-lesson connections — affirm these publicly.
Model Building Phase 📹 **VIDEO:** More functional groups Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chargaff's Base-Pairing Rules - Advanced (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the cumulative explanatory model begun in Lesson 1 (a diagram showing salt types and the three phenomenon salts). They add a Solubility column to their model, labelling NaCl as soluble, Na_2CO_3 as soluble and NH_4NO_3 as soluble — and noting that solubility in liquid water does not yet explain the overnight observations (because no liquid water was present). They draw a dashed arrow from the NH_4NO_3 box labelled 'Why does it clump if there is no liquid water?' to signal an unresolved question within the model itself. In the final two minutes, learners do a silent gallery walk of two neighbouring models and add one green dot sticker if they agree with the solubility labels or one orange dot if they see an error.	Say: 'Open your model from Lesson 1. You are going to add one new layer — solubility. But here is the challenge: if NaCl, Na_2CO_3 and NH_4NO_3 are all soluble, and there was no liquid water in the storeroom, your model should show that we still cannot fully explain the phenomenon. Scientists call that an open question — show it with a dashed arrow.' Circulate during model building. If a learner labels CAN as insoluble, crouch and ask: 'What did your test tube show you for ammonium nitrate?' WAIT TIME: 5 seconds. During the gallery walk, say: 'You are not looking for a perfect model — you are looking for one thing you could argue with and one thing you agree with.'	The cumulative model is the central sensemaking artefact of the unit. Adding a solubility layer while explicitly marking what remains unexplained models the scientific practice of representing both knowledge and uncertainty. The silent gallery walk with dot stickers is a low-stakes peer assessment that builds community norms around constructive critique.	Collect or photograph two to three models at the end of the lesson. Check that learners have: (1) added correct solubility labels for all three phenomenon salts, (2) included a dashed arrow or question mark for the unresolved clumping question, and (3) not prematurely explained the phenomenon using ideas not yet taught (e.g. some learners may write hygroscopic — if so, note their model as an advanced starting point for Lesson 6 discussion).

D. TEACHER REFLECTION
Ask yourself: Did learners move from predicting based on appearance to predicting based on salt type by the end of the lesson? Did the class table reveal any experimental errors (e.g. too much salt added, causing all results to appear insoluble) that need addressing before Lesson 3? Were learners able to articulate why solubility in liquid water alone does not explain the overnight phenomenon — or did they conflate liquid water with water vapour? If the majority of orange DQB questions do not mention water vapour or air moisture, plan a brief provocation at the start of Lesson 3: ask learners to smell the air in the storeroom compared to outside and consider what is different. Also reflect: were Kiswahili key terms on the reading card sufficient for learners who are not confident in English scientific vocabulary? If not, prepare a bilingual glossary card for Lesson 3.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	When ammonium nitrate (the salt in CAN fertiliser), sodium chloride (table salt) and sodium carbonate (washing soda) were added to water in test tubes, all three dissolved — the mixtures became clear solutions. Calcium carbonate remained as a cloudy suspension, indicating it did not dissolve. Copper sulphate dissolved and produced a blue solution. Lead sulphate remained cloudy and insoluble.
What did I learn?	Solubility depends on the type of salt, not its colour or appearance. All nitrates are soluble. Most chlorides are soluble. Most carbonates are insoluble except those of sodium, potassium and ammonium. Most sulphates are soluble except lead(II) sulphate and barium sulphate. These rules form the beginning of a solubility reference chart we will use throughout the unit.
How does this explain the phenomenon?	We can now explain why table salt dissolves in githeri water (NaCl is soluble) and why calcium carbonate does not dissolve in river water (it is insoluble). However, we cannot yet explain why CAN fertiliser clumped overnight in the storeroom — because there was no liquid water present, only air. This means solubility in liquid water is not the full explanation. Something about the air itself must be involved, and that is what we will investigate in Lessons 3 to 6.

LESSON 3 (80 minutes): Preparing Soluble Salts I — Acid + Metal and Acid + Metal Oxide (Experiment 2a)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners prepare zinc sulphate ($ZnSO_4$) by reacting zinc metal with dilute sulphuric acid, and copper sulphate ($CuSO_4$) by reacting copper(II) oxide with dilute sulphuric acid. They collect and test the hydrogen gas produced in the acid-metal reaction, write balanced chemical equations for both reactions, and connect their laboratory observations to the anchoring phenomenon — explaining why CAN fertiliser (which contains soluble salts formed through acid-base and acid-metal reactions) absorbs moisture from the air.
Knowledge	a) A salt is formed when a hydrogen atom in an acid is fully or partially replaced by a positive ion (metal ion or ammonium ion). b) Dilute sulphuric acid reacts with zinc metal to produce zinc sulphate and hydrogen gas: $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$. c) Dilute sulphuric acid reacts with copper(II) oxide to produce copper sulphate and water: $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$. d) Hydrogen gas is identified by the 'pop' test using a burning splint. e) The preparation method chosen depends on the solubility of the salt — soluble salts such as $ZnSO_4$ and $CuSO_4$ are prepared by acid + metal or acid + metal oxide routes.
Skills	b) Prepare salts using appropriate methods in the laboratory. The learner carries out: acid + metal ($Zn + H_2SO_4$) and acid + metal oxide ($CuO + H_2SO_4$) experiments; collects and tests hydrogen gas; filters and evaporates to obtain crystalline product; writes and balances chemical equations; records observations systematically.
Attitudes	e) Appreciate the applications of salts in day-to-day life. Learners develop responsibility by taking care of apparatus and observing safety when handling dilute sulphuric acid. They develop teamwork through group decision-making during the experiment and collaborate to share evidence on the Driving Question Board.
Key Inquiry Question	How can salts be prepared in the laboratory?
Purpose in Storyline	In Lessons 1–2, learners classified salts (normal, acidic, basic, double) and investigated solubility patterns (all sulphates are soluble except $BaSO_4$, $PbSO_4$, and slightly soluble $CaSO_4$). They now know that CAN fertiliser contains calcium ammonium nitrate — a soluble salt. Lesson 3 advances the storyline by showing HOW soluble sulphate salts are made in the laboratory: learners prepare $ZnSO_4$ (acid + metal) and $CuSO_4$ (acid + metal oxide), write balanced equations, and collect the first direct experimental evidence that acid-salt reactions are the same family of reactions that produced the fertiliser granules in their storeroom — explaining why those granules are hygroscopic and clump when exposed to humid Kenyan air.
Safety Notes	Environmental issues (safety in the laboratory): The learner observes safety when handling acids and bases. Specific precautions: (1) Dilute H_2SO_4 is corrosive — learners must wear safety goggles and gloves at all times; spills must be neutralised with sodium hydrogen carbonate solution before disposal. (2) Hydrogen gas is flammable — all naked flames must be extinguished before collecting gas; re-light splint only when instructed. (3) Excess zinc and copper oxide must be collected in a labelled waste container and disposed of according to school laboratory waste protocols — not washed into the drainage system. (4) Hot filtrate must be handled using heat-resistant gloves. (5) Learners with skin sensitivity must report to the teacher before handling reagents.

B. LESSON OVERVIEW

Learners work in groups to carry out two linked experiments: (1) preparing zinc sulphate by reacting zinc granules with dilute sulphuric acid, with collection and testing of hydrogen gas by the 'pop' test; and (2) preparing copper sulphate by reacting black copper(II) oxide powder with dilute sulphuric acid, observing the disappearance of the black solid and formation of a blue solution. Both products are filtered and the filtrate retained for evaporation. Learners write balanced chemical equations for both reactions and connect their observations to a short teacher-facilitated video demonstration of $ZnSO_4$ preparation. The lesson anchors constantly to the driving-question

phenomenon: the CAN fertiliser granules in the Rift Valley storeroom are soluble salts produced through acid reactions — and understanding how they are made is the first step to explaining why they clump and absorb moisture from the air. The DQB is updated with new mechanistic evidence: 'Salts like $ZnSO_4$ and $CuSO_4$ are made when acids react with metals or metal oxides — the same family of reactions that made CAN fertiliser.'

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Acid Rain (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually complete a 'Predict-Record' card (provided on ARES). The card shows two setups: Setup A — a beaker containing zinc granules with dilute sulphuric acid being poured in; Setup B — a beaker containing black copper(II) oxide powder with dilute sulphuric acid being poured in. Learners write predictions for: (1) What will you observe in Setup A? (2) What will you observe in Setup B? (3) What product(s) do you expect to form in each case? (4) How does this experiment connect to what happened to the CAN fertiliser bag in the Rift Valley storeroom? Learners are given 4 minutes to write silently, then share predictions with a shoulder partner for 2 minutes. The teacher cold-calls 4 learners — 2 for Setup A, 2 for Setup B — to share predictions aloud and records key prediction phrases on the board in two columns headed 'Setup A predictions' and 'Setup B predictions'. These columns remain visible throughout the lesson.	Teacher opens by displaying the phenomenon image on ARES (the clumped CAN fertiliser bag) and says: 'Before we touch any equipment today, I want you to think like a chemist. Look at these two setups on your screen.' [Display Setup A and Setup B diagrams on ARES.] 'Take 4 minutes of silent thinking and writing — no talking yet.' [Start 4-minute timer. Walk the room, do NOT give hints, observe what learners write.] After 4 minutes: 'Now turn to your shoulder partner. You have 2 minutes — share your prediction and explain your reasoning.' [Start 2-minute timer.] After sharing: 'I am going to cold-call four people. I want your PREDICTION — not the right answer.' [Wait 10 seconds.] Cold-call Learner 1 (Setup A): 'What do you predict will happen when zinc meets sulphuric acid?' Record key words verbatim. Cold-call Learner 2 (Setup A, different prediction): 'Does anyone predict something different for Setup A?' Cold-call Learner 3 (Setup B): 'What do you predict for the black copper oxide and acid?' Cold-call Learner 4 (Setup B, different view): 'Interesting — anyone predict differently for Setup B?' Teacher closes phase: 'Excellent. We now have our predictions on the board. Keep them there — by the end of	Predict-Record-Share (Elicit Prior Knowledge Strategy): By committing predictions to writing BEFORE observation, learners activate prior knowledge from Lessons 1–2 (acid properties, salt formation) and create a cognitive anchor point. Recording predictions publicly on the board creates accountability — learners are motivated to compare their predictions against evidence, which is the core mechanism of conceptual change. Connecting the prediction explicitly to the CAN phenomenon ensures the lesson does not feel like an isolated lab exercise.	Teacher scans written Predict cards during the silent writing phase. Observable indicators of readiness: (a) Learner correctly predicts gas production in Setup A (demonstrates recall of acid + metal = salt + hydrogen from Lesson 2 or prior knowledge). (b) Learner predicts colour change or dissolution of black solid in Setup B (demonstrates understanding of metal oxide + acid). (c) Learner makes any connection — even tentative — between the setups and the CAN fertiliser phenomenon. Learners who predict 'nothing will happen' or leave the phenomenon connection blank are flagged for targeted questioning during the Observe phase.

		today you will know which ones to tick, which to revise, and why.'		
Observe Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Acid Rain (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of four. Each group collects an equipment tray pre-set by the teacher. EXPERIMENT A (Acid + Metal — $ZnSO_4$ preparation): Learners add ~2 g zinc granules to a conical flask, then carefully add 20 cm^3 dilute H_2SO_4 (1 mol/dm^3). They observe gas production immediately. Using a delivery tube and inverted test tube filled with water (water displacement), learners collect a test tube of gas. They test the gas with a burning splint (pop test) to confirm H_2. They continue adding excess zinc until the reaction stops (ensuring all acid is consumed — a critical technique for obtaining a pure salt). They filter the mixture to remove unreacted zinc and collect the clear $ZnSO_4$ filtrate. EXPERIMENT B (Acid + Metal Oxide — $CuSO_4$ preparation): Learners add ~2 g black CuO powder to a beaker, then carefully add 20 cm^3 dilute H_2SO_4 (1 mol/dm^3) and warm gently on a tripod/gauze. They stir and observe the black powder dissolving and the solution turning blue. Excess CuO is added (to ensure all acid is consumed) until some black powder remains undissolved. The mixture is filtered to remove excess CuO. The blue $CuSO_4$ filtrate is collected. Throughout both experiments, learners record observations in a structured table on ARES: columns for 'What I saw', 'What I heard/smelled', 'What I measured', 'Connection to CAN phenomenon'.	Teacher distributes equipment trays and says: 'Before you touch anything, read the safety card on your tray. Who can tell me the THREE safety rules for today?' [Wait 10 seconds. Cold-call 1 learner.] Confirm rules aloud: goggles on, no naked flames during gas collection, acid spills → sodium hydrogen carbonate. 'Now — Experiment A first. Follow the ARES procedure step by step. Your job is to be a scientist: record EXACTLY what you see, hear, and smell at each step.' [Start Experiment A. Circulate.] During Experiment A, stop the class after gas collection: 'EVERYONE PAUSE. Before you do the pop test, extinguish all flames on your bench. Thumbs up when your flame is out.' [Confirm all flames out.] 'Now — one person per group lights the splint and tests the gas. What do you hear?' [Allow pop tests. Circulate, listen for 'pop' reports.] 'What does the pop tell us? Turn and tell your partner — 10 seconds.' [Wait 10 seconds.] Cold-call 2 learners: 'What gas did we just confirm?' After Experiment A filtration, transition: 'Now Experiment B. Notice the colour of the CuO powder before you start — write it down. What do you predict the solution will look like after you add acid and warm it?' [Wait 10 seconds.] [Start Experiment B. Circulate, asking probing questions: 'Why are we adding EXCESS CuO?', 'What would happen to our salt if we left unreacted acid in the solution?'] After both experiments: 'Complete your observation table on ARES,	Structured Observation with Phenomenon Anchoring: The ARES observation table forces learners to explicitly record a 'Connection to CAN phenomenon' column for every observation. This is not decorative — it prevents the experiment from becoming a disconnected procedure and constantly asks learners: 'What does this tell you about the fertiliser?' The excess-reagent technique (adding excess zinc / excess CuO) is introduced here as a purposeful decision linked to purity of the salt product — this directly connects to why industrial fertiliser manufacturers must ensure complete acid consumption to produce stable, pure salt granules.	Teacher monitors three observable indicators during the Observe phase: (1) SAFETY COMPLIANCE — all groups wearing goggles, flames extinguished before pop test. Any non-compliance triggers immediate whole-class pause and re-instruction. (2) TECHNIQUE — groups add excess zinc/CuO correctly; any group adding acid in excess is redirected with the question: 'What impurity will be in your filtrate if you have leftover acid?' (3) OBSERVATION QUALITY — teacher spot-checks ARES observation tables; looks specifically for whether learners record the COLOUR CHANGE in Experiment B (colourless/white CuO → blue solution) and whether the GAS TEST result is correctly linked to H_2 identity. Groups who only write 'bubbles formed' without identifying the gas are prompted: 'What did the pop test tell you about the identity of the gas?'

		including the CAN phenomenon column. You have 3 minutes.' [Timer. Walk the room.]		
Explain Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Acid Rain (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners transition from observation to explanation in three steps. STEP 1 — Video anchor (3 min): The ARES platform plays a short teacher-narrated demonstration video of $ZnSO_4$ preparation, showing close-up of gas collection, the pop test, and the clear filtrate. Learners compare what they saw in their own experiment to the video, noting any differences. STEP 2 — Equation writing (10 min): Working in pairs, learners write and balance the two chemical equations on ARES: Reaction A: $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$. Reaction B: $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$. Learners must: (a) include correct state symbols; (b) confirm that equations are balanced (atom count); (c) label which reactant is the acid and which is the metal / metal oxide; (d) identify the salt product in each equation and state its name. STEP 3 — Phenomenon connection (7 min): Learners respond individually on ARES to the prompt: 'CAN fertiliser (calcium ammonium nitrate) is a soluble salt. Using what you have learned today about how soluble salts are prepared, write 2–3 sentences explaining how a salt like CAN could be produced through an acid reaction. Then explain: why does knowing how CAN was made help us understand why it clumps in the storeroom?' Learners share responses with their group and refine their answers before the DQB phase.	Teacher opens Explain phase: 'Experiments done. Now we move from WHAT happened to WHY it happened.' [Play $ZnSO_4$ video on ARES, 3 min.] After video: 'One difference and one similarity between your experiment and the video — write it on your ARES note. 30 seconds.' [Wait 30 seconds.] 'Now equation time. In your pairs, write Reaction A first. I want to see: correct formulae, balanced, state symbols, product labelled as a salt. You have 4 minutes.' [Timer. Circulate. Look for common errors: omitting state symbols, forgetting H_2 on the product side, writing H_2SO_4 as H_2SO_3.] After 4 minutes: 'Stop. Cold-call — ' [Name Learner A] ' — read me your balanced equation for Reaction A.' [Learner reads.] 'Class — do you agree? Is it balanced? Count the atoms with me.' [Teacher counts aloud with class: Zn: 1=1 ✓, H: 2=2 ✓, S: 1=1 ✓, O: 4=4 ✓.] Repeat for Reaction B. 'Now the big connection — 7 minutes to answer the CAN question on ARES individually, then share with your group.' [Timer. Circulate, read responses on ARES dashboard.] After sharing: 'I saw some powerful ideas. ' [Name Learner B] ', share what your group said about why knowing HOW CAN is made helps explain why it clumps.' [Wait for response.] 'Excellent. That is exactly what we are going to capture on our Driving Question Board now.'	Equation-to-Phenomenon Bridge (Representational Mapping Strategy): Learners move through three levels of representation — (1) macroscopic observation (what they saw in the experiment), (2) symbolic representation (balanced equations with state symbols), and (3) phenomenon-level explanation (connecting the chemistry of salt preparation to the behaviour of CAN fertiliser in a Kenyan storeroom). This three-level mapping, structured by the ARES prompts, builds scientific explanatory reasoning — the core NGSS practice of Constructing Explanations. The pair equation-writing before cold-call sharing ensures all learners attempt the symbolic level before receiving peer/teacher feedback.	EQUATION CHECK: Teacher scans all ARES-submitted equations before cold-calling. Key error indicators: (a) Unbalanced equations — teacher asks the class to 'count atoms together' rather than simply correcting, preserving reasoning practice. (b) Missing state symbols — teacher asks: 'Is $ZnSO_4$ a solid or in solution? How do we know?' (c) Wrong products — e.g., writing ZnO instead of $ZnSO_4$ — teacher asks: 'What is the definition of a salt? Does ZnO fit that definition?' PHENOMENON RESPONSE CHECK: Observable indicator of mastery — learner explicitly states that CAN is a soluble salt produced through an acid reaction, AND connects solubility to moisture absorption. Learners who only describe the reaction without connecting to the phenomenon are flagged for DQB phase follow-up.

Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Acid Rain (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each group receives two DQB sticky note colours (provided on ARES as digital cards): BLUE = 'New evidence we gathered today' and GREEN = 'New questions this raises'. Groups collaboratively compose: (1) ONE blue card summarising the key mechanism evidence from today: e.g., 'Soluble salts like $ZnSO_4$ and $CuSO_4$ can be prepared by reacting an acid with a metal or metal oxide — the acid's hydrogen is replaced by a metal ion to form a salt.' (2) ONE green card raising a new question prompted by today's evidence: e.g., 'If CAN fertiliser was made using an acid reaction, what was the acid and what was the metal/base that reacted?' or 'What happens to $ZnSO_4$ and $CuSO_4$ if we leave them in the Kenyan humidity overnight — will they also clump like CAN?' Groups post their cards on the class DQB (physical or ARES digital board). The teacher reads two blue cards and two green cards aloud, connecting them explicitly to the driving question displayed at the top of the DQB: 'Why did the fertiliser granules get wet and clump together…?'	Teacher transitions: 'Time to update our board. Remember — this board is tracking our growing understanding of the big question about the fertiliser storeroom.' [Point to or display the driving question.] 'Your group has 3 minutes to write your blue card — one sentence capturing the most important new evidence from today's experiments.' [Timer. Walk room, prompt groups who are stuck: 'What did you prove about HOW the salt was formed? What happened to the acid's hydrogen?'] After 3 minutes: 'Now your green card — one question that today's evidence raised for you. It must connect to the driving question. 2 minutes.' [Timer.] After posting: 'Let me read some of these out.' [Read 2 blue cards.] 'Notice — both of these blue cards say the salt is formed when a metal ion REPLACES hydrogen in the acid. That is our mechanism. That is new knowledge.' [Read 2 green cards.] 'These green questions are exactly what the rest of our lesson sequence will answer. Keep them here. We will come back to them.' Close: 'By the end of the 8 lessons, every green card on this board should be answered by a blue card. That is how science builds.'	DQB as Cumulative Evidence Tracker (Argumentation and Evidence Mapping Strategy): The two-colour DQB card system (blue = evidence, green = new questions) makes the progressive structure of scientific knowledge-building visible and tangible. Requiring groups to write in complete sentences — not just keywords — forces learners to articulate the causal mechanism (acid + metal/metal oxide → salt) rather than simply naming the reaction type. Connecting every card to the driving question prevents the DQB from becoming a passive display and keeps the phenomenon as the unit's organising thread.	Blue card quality check — Observable indicator of mastery: the card explicitly names the mechanism (H in acid replaced by metal ion) rather than merely describing the observation ('zinc dissolved'). If groups write only observational statements, teacher prompts: 'What CHEMICALLY caused the zinc to dissolve? What was happening to the acid's hydrogen atoms?' Green card quality check — indicator of productive curiosity: the question is answerable by further chemistry investigation (not a trivial observation question). Teacher notes any green questions that reveal misconceptions (e.g., 'Did we make CAN fertiliser today?') to address in the next lesson.
Model Building Phase 📹 **VIDEO:** Video: pH Scale Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos	Learners return to the cumulative class model that was started in Lesson 1 and updated in Lesson 2. The model is displayed on ARES as a shared diagram. In Lesson 1, the model showed the three substances from the phenomenon (CAN, washing soda, table salt) with question marks around their behaviour. In Lesson 2, the model added the	Teacher opens model phase: 'Final task. Open your cumulative model on ARES. You built this in Lesson 1. You updated it in Lesson 2. Today we add the preparation pathways.' [Display the shared class model on the board.] 'Look at the soluble salt box in your model. Today you proved two ways to get there — acid plus metal, and acid plus metal oxide.	Cumulative Model Revision (NGSS Developing and Using Models Practice): The layered, cumulative model is the central sensemaking artefact of the entire 8-lesson unit. By returning to the SAME model every lesson and adding a new explanatory layer, learners experience science as a progressive, evidence-driven process of model refinement —	Model check — Observable indicators of mastery: (a) BOTH preparation pathway arrows present (acid + metal AND acid + metal oxide), each with a correctly balanced example equation. Learners with only one arrow are prompted: 'You did TWO experiments today — where is the second pathway?' (b) Speech bubble contains both a causal

📖 **HTML:** [Acid Rain (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	classification layer (normal salt, acidic salt, soluble/insoluble). Today, learners add a PREPARATION PATHWAY layer to the model. Working individually, each learner draws or annotates (on their ARES personal copy) two new arrows entering the 'soluble salt' box: Arrow 1: Acid + Metal → Soluble Salt + Hydrogen gas (example: $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$). Arrow 2: Acid + Metal Oxide → Soluble Salt + Water (example: $CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$). Learners add a speech bubble to the CAN fertiliser bag image in the model: 'I was made using an acid reaction — that is why I am a soluble salt, and soluble salts can absorb water from the air.' Learners compare their updated model with their Lesson 1 model and write one sentence on ARES: 'What does my model now explain that it could not explain before Lesson 3?'	Add both arrows with their equations. You have 5 minutes.' [Timer. Circulate. Check that learners are adding BOTH arrows with equations — not just one.] After 5 minutes: 'Now add the speech bubble to the CAN bag. What would the CAN bag say if it understood its own chemistry?' [Wait 10 seconds.] Cold-call 2 learners to read their speech bubbles aloud. [Listen for: soluble salt + acid reaction connection + water absorption link.] Final reflection: 'Write one sentence: What does your model now explain that it couldn't in Lesson 1? Post it on ARES. You have 90 seconds.' [Timer.] Close the lesson: 'In Lesson 4, we are going to prepare salts using a completely different method — acid plus carbonate and acid plus hydrogen carbonate. By then, your model will have four preparation pathways, and we will be much closer to explaining everything about that fertiliser bag.'	not as a series of disconnected topics. The speech bubble technique (giving voice to the CAN fertiliser bag) is a low-stakes creative strategy that requires learners to synthesise the lesson's chemistry into a phenomenon-level explanation, making the mechanism personally meaningful. The 'what does my model now explain' reflection sentence makes learning gains explicit and metacognitive.	mechanism (soluble salt, acid reaction) AND a connection to atmospheric behaviour (water absorption / clumping). (c) Reflection sentence demonstrates awareness of model growth: e.g., 'My model can now explain WHERE the salt in CAN fertiliser came from, not just what type of salt it is.' Learners whose reflection sentences remain at the observational level (e.g., 'My model now shows two experiments') are identified for targeted review at the start of Lesson 4.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions to inform planning for Lesson 4 (Acid + Carbonate / Acid + Hydrogen Carbonate):

1. SAFETY EXECUTION: Did all groups extinguish flames before the pop test? If any group did not, what adjustment to the safety briefing is needed — consider assigning a dedicated 'safety monitor' role within each group in Lesson 4.

2. EQUATION QUALITY: Review the ARES-submitted equations. What were the most common errors — missing state symbols, unbalanced equations, or wrong products? Plan a 3-minute targeted correction at the start of Lesson 4 using one anonymised 'almost-correct' equation as a class editing exercise.

3. PHENOMENON CONNECTION: Check the DQB green cards and model speech bubbles. Did learners make the explicit link between soluble salt preparation (acid + metal/metal oxide) and the hygroscopic behaviour of CAN fertiliser? If fewer than 70% of learners made this connection explicitly, re-open the phenomenon image at the start of Lesson 4 and ask: 'We now know CAN is made through an acid reaction — so what property of a soluble salt explains the clumping?'

4. EXCESS REAGENT UNDERSTANDING: Did learners understand WHY excess zinc/CuO is added? This technique is critical for Lesson 4 (acid + carbonate) and Lesson 5 (precipitation). If groups struggled with this concept, plan a 2-minute targeted question at the start of Lesson 4: 'What would happen to the purity of our $CuSO_4$ if we had leftover acid in the filtrate?'

5. GROUP DYNAMICS: Note any groups where one learner dominated the experiment. In Lesson 4, rotate roles explicitly (recorder, equipment handler, safety monitor, reporter) to ensure Communication and Collaboration competency is developed across all learners.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Zinc granules reacted with dilute sulphuric acid to produce a colourless solution and a gas that caused a burning splint to 'pop'. Black copper(II) oxide powder dissolved in warm dilute sulphuric acid to produce a blue solution. In both cases, the solid was added in excess and the unreacted solid was removed by filtration, leaving a clear salt solution.
What did I learn?	A salt is formed when the hydrogen in an acid is replaced by a metal ion. Zinc sulphate ($ZnSO_4$) is prepared by the acid + metal reaction: $Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$. Copper sulphate ($CuSO_4$) is prepared by the acid + metal oxide reaction: $CuO(s) + H_2SO_4(aq) \rightarrow CuSO_4(aq) + H_2O(l)$. Both $ZnSO_4$ and $CuSO_4$ are soluble salts — all sulphates are soluble except $BaSO_4$, $PbSO_4$, and slightly soluble $CaSO_4$. Excess metal or metal oxide is added to ensure all the acid is consumed, giving a pure salt product. Hydrogen gas is identified by the 'pop' test with a burning splint. The preparation method used for a salt depends on its solubility.
How does this explain the phenomenon?	CAN fertiliser (calcium ammonium nitrate) is a soluble salt produced through acid reactions — the same family of reactions we carried out in the laboratory today. Because CAN is a soluble salt, its granules can interact with water molecules in humid air. Now that we understand HOW soluble salts like CAN are prepared (acid replaces its hydrogen with a metal ion to form a soluble salt), we are one step closer to explaining why those soluble salt granules absorb moisture from the air and clump together in the Rift Valley storeroom — while table salt (NaCl) and washing soda behave differently. Our next question is: how are salts prepared using carbonates and hydrogen carbonates — and will those methods add more evidence to our explanation?

LESSON 4 (80 minutes): Preparing Soluble Salts II — Acid + Base and Acid + Carbonate/Hydrogen Carbonate	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners prepare sodium chloride via neutralisation (HCl + NaOH) and calcium chloride via acid + carbonate reaction (HCl + $CaCO_3$), collect and test CO_2 gas, write full balanced equations, and compare all four soluble salt preparation methods using a summary flow chart.
Knowledge	a) Salts can be prepared by reacting an acid with a base (neutralisation) or with a carbonate/hydrogen carbonate; b) The acid + carbonate/hydrogen carbonate reaction produces CO_2 gas, water, and a salt; c) Balanced chemical equations represent conservation of mass in salt preparation reactions; d) The choice of preparation method depends on the solubility and reactivity of the salt being formed.
Skills	b) Prepare salts using appropriate methods in the laboratory; b) Write balanced chemical equations for reactions involved in the preparation of salts; b) Collect and test for gases produced during experiments (CO_2 with limewater); b) Properly dispose of waste after experiments.
Attitudes	e) Appreciate the applications of salts in day-to-day life, including their role in agriculture (CAN fertiliser) and food preparation (table salt/baking soda in mandazi); Responsibility — the learner develops accountability as they take care of apparatus while using them during experiments.
Key Inquiry Question	How can salts be prepared in the laboratory?
Purpose in Storyline	In Lesson 3, learners prepared soluble salts via acid + metal and acid + metal oxide reactions (Experiment 2a). Today (Experiment 2b), learners advance the investigation by using two more preparation routes — neutralisation (acid + base) and acid + carbonate — producing NaCl and $CaCl_2$ respectively. The CO_2 gas produced in the carbonate reaction directly echoes the fizzing observed when baking soda (sodium hydrogen carbonate) is added to mandazi batter, connecting chemistry to a beloved Kenyan snack. By completing a four-method summary flow chart, learners now have the full toolkit of soluble salt preparation methods needed to later explain why CAN fertiliser absorbs moisture from the air and why understanding salt preparation matters for safe, effective use in Kenyan agriculture.
Safety Notes	Environmental issues (safety in the laboratory — PCI): HCl and NaOH are corrosive — learners must wear safety goggles and gloves at all times. No mouth pipetting. Acid must be added to water, never the reverse. NaOH pellets/solution must not contact skin; wash immediately with water if contact occurs. Limewater ($Ca(OH)_2$) is an irritant — avoid skin contact. All waste solutions must be neutralised before disposal down the sink with excess water. Broken glassware must be reported to the teacher and disposed of in the designated sharps bin, not the general waste bin. Learners with asthma should be positioned away from CO_2 gas collection. Teacher must inspect all apparatus for cracks before the experiment begins.

B. LESSON OVERVIEW

This is Lesson 4 of 8 in the evidence-gathering sequence for Sub-Strand 2.1: Introduction to Salts. Building directly on Lesson 3 (acid + metal and acid + metal oxide), learners now carry out Experiment 2b — two new soluble salt preparation routes. In Part A, they prepare sodium chloride (NaCl) by neutralising hydrochloric acid with sodium hydroxide solution, using universal indicator to identify the endpoint. In Part B, they prepare calcium chloride ($CaCl_2$) by reacting hydrochloric acid with calcium carbonate (marble chips/limestone — abundant in Kenya's Coast and Central regions), collecting the CO_2 gas produced and testing it with limewater. The CO_2 observation is linked to the mandazi supporting phenomenon (baking soda fizzing in dough) and to the anchoring phenomenon (washing soda efflorescence). Learners write and balance full chemical equations for both reactions. The lesson culminates in the construction of a four-method summary flow chart comparing all soluble salt preparation routes studied so far (direct synthesis, acid + metal, acid + metal oxide, acid + base, acid + carbonate). This flow chart is added to the Driving Question Board as a growing evidence artefact. The lesson addresses KICD Core Competencies: Communication and Collaboration (group experiment and discussion) and

Digital Literacy (optional: searching applications of acids and bases). PCIs addressed: Environmental Issues / Laboratory Safety.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually write a prediction in their science notebooks in response to two displayed questions: (1) 'When HCl reacts with NaOH solution, what products do you think will form? Will you be able to see any change happening?' and (2) 'When HCl reacts with marble chips ($CaCO_3$), what do you predict will happen — and how is this similar to what happens when you add baking soda to mandazi dough?' Learners then share predictions with a partner (Think-Pair-Share) before three pairs are cold-called to share with the class. Predictions are recorded on a sticky-note section of the Driving Question Board labelled 'Our predictions for Experiment 2b'. Learners also revisit their Lesson 3 notes to identify the pattern of acid + metal oxide → salt + water and predict whether a similar pattern exists for acid + base.	Display both prediction questions on the board. Say: 'Before we touch any apparatus today, I want you to think like a scientist — make a prediction first. Write your prediction in your notebook. You have 2 minutes of silent thinking time.' [Allow 2 minutes silent individual writing.] After writing, say: 'Now turn to your neighbour and compare your predictions. Do you agree? Do you disagree? You have 90 seconds.' [Allow 90 seconds pair discussion.] After pair discussion, cold-call 3 pairs — deliberately select one pair who likely predicted no visible change for neutralisation, one who predicted fizzing for carbonate, and one who mentioned mandazi. Use exact phrases: 'Wanjiru and Kamau, what did your pair predict for the neutralisation reaction?' [10-second wait time after each cold-call before accepting answer.] After three pairs share, say: 'Notice that some of us predicted fizzing for the carbonate reaction — where have you seen fizzing like that in real life in Kenya? Think about your kitchen.' Accept 2–3 rapid responses. Bridge: 'Keep your predictions visible — by the end of today, you will know if you were right, and more importantly, WHY.' Post class predictions on sticky notes on the DQB.	Think-Pair-Share with Anchoring Phenomenon Connection — Learners activate prior knowledge of acid reactions (Lessons 2 & 3) and connect to the mandazi supporting phenomenon (CO_2 from $NaHCO_3$) before observing. This primes pattern recognition: acid + carbonate → CO_2, linking kitchen chemistry to laboratory chemistry and building conceptual continuity across the storyline.	Teacher circulates during silent writing and reads 4–5 notebooks. Observable indicator of readiness: learner can write at least one reactant and one predicted product for each reaction (e.g., 'HCl + NaOH → salt + water'). If fewer than half the class can articulate a product, teacher pauses and asks: 'In Lesson 3, acid + metal oxide gave us salt + ___? [water]. So what might acid + base give?' before proceeding.
Observe Phase	Learners work in groups of 4. Each group receives a pre-labelled	Before distributing equipment, say: 'Before we start — safety check.	Structured Observation with a Results Table — A pre-formatted	Teacher checks each group's results table at the 22-minute

📹 **VIDEO:** [Precipitation reactions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	equipment tray. The experiment has two parts run sequentially. PART A — Neutralisation (NaCl preparation): Learners measure 25 cm^3 of 1 mol/dm^3 HCl into a conical flask and add 3 drops of universal indicator. They record the colour and pH. They then add 1 mol/dm^3 NaOH solution from a burette or dropper, swirling after each addition, recording colour changes in a results table. They identify the endpoint (green colour, pH ≈ 7) and note: 'The solution looks like water — there is no visible precipitate.' PART B — Acid + Carbonate ($CaCl_2$ preparation, with CO_2 gas collection): Learners add a spatula of marble chips ($CaCO_3$, labelled 'limestone — from Coast region, Kenya') to a boiling tube. They connect a delivery tube leading into a test tube of limewater. They add dilute HCl (2 mol/dm^3) through a thistle funnel. Learners observe: (a) fizzing/effervescence in the boiling tube, (b) limewater turning milky/cloudy — confirming CO_2. They record all observations in a structured table with columns: Reactants | Observation during reaction | Observation of limewater | Inference. Groups photograph their limewater result using a tablet (if available) for the DQB. Learners compare the fizzing to the mandazi baking soda observation from the lesson 1 phenomenon walk.	Everyone: goggles on, gloves on. Show me.' [Visually confirm all 30 learners are wearing PPE before proceeding.] Distribute equipment trays. Say: 'You will find the step-by-step instruction card in your tray. Your group has exactly 25 minutes to complete both parts. The roles today are: Apparatus Handler, Recorder, Safety Monitor, and Reporter. Assign roles now — you have 30 seconds.' [Allow role assignment.] Circulate continuously. At the 8-minute mark, check on Part A: ask groups, 'What colour is your indicator now? What does that tell you about the pH?' [10-second wait time.] At the 15-minute mark, check Part B CO_2 collection: if limewater has not turned milky, ask: 'Is your delivery tube properly submerged in the limewater? Check the seal at the bung.' For groups that have milky limewater, ask: 'What gas caused this change? How do you know?' [10-second wait time.] At the 22-minute mark, prompt all groups: 'Recorders — is your results table complete for both parts? Safety monitors — are all acid bottles capped?' With 3 minutes remaining, say: 'Begin cleaning up according to the waste disposal procedure on your card. Neutralise your waste solutions before pouring down the sink.' Collect all sharps/broken glass reports. After clean-up, bring the class together: cold-call 2 groups to share Part A observations and 2 groups to share Part B observations (4 cold-calls total, 10-second wait time each).	results table (Reactants | Observation during reaction | Limewater observation | Inference) scaffolds learners to move from raw sensory observation to scientific inference. This is NGSS SEP 3 (Planning and Carrying Out Investigations) and SEP 4 (Analysing and Interpreting Data). The CO_2 → limewater milky link makes the invisible gas (CO_2) visible, directly echoing how CO_2 makes mandazi rise and how CO_2 is released from washing soda (Na_2CO_3) when it reacts with atmospheric moisture — a thread back to the anchoring phenomenon.	mark. Observable indicators of success: (1) Part A — learner records the colour progression of universal indicator (red → orange → yellow → green) and identifies pH ≈ 7 as the endpoint; (2) Part B — learner records 'fizzing/bubbles in boiling tube' AND 'limewater turned milky/cloudy' and infers 'CO_2 gas was produced'. Red flag: if a group records no change in limewater, teacher checks equipment setup and asks the group to repeat the gas test before moving on.
Explain Phase 📹 **VIDEO:**	Learners first complete a 'Word Equation → Balanced Equation'	Display the word equation scaffold on the board. Say: 'Every reaction	Word Equation → Symbol Equation Scaffold with Atom	Teacher collects 6 notebooks (one per group's recorder) at the end of

[Precipitation reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	scaffold in their notebooks for both reactions: (A) Hydrochloric acid + Sodium hydroxide → Sodium chloride + Water; (B) Hydrochloric acid + Calcium carbonate → Calcium chloride + Carbon dioxide + Water. They then write the full balanced symbol equations: (A) $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$; (B) $2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$. Learners label state symbols and identify the type of salt formed in each case. They then complete a 'Phenomenon Connection' box: 'The CO_2 produced in Reaction B is the same gas that makes mandazi dough rise when baking soda ($NaHCO_3$) reacts with the acid in the dough. Write the equation for that reaction too: $2HCl + NaHCO_3 \rightarrow NaCl + CO_2 + H_2O$.' As a group, learners compare Reactions A and B: 'In which reaction did you see gas? In which reaction did you NOT see a visible change? How do you know a reaction actually happened in neutralisation?' Groups discuss and share answers. Learners also identify that $CaCl_2$ (calcium chloride) is a hygroscopic salt — linking to the anchoring phenomenon (CAN fertiliser clumping — CAN contains ammonium nitrate and calcium compounds).	we observed today follows a pattern. Your job now is to translate what you saw into the language of chemistry — balanced equations. Start with the word equation, then convert. You have 5 minutes individually.' [Allow 5 minutes.] After 5 minutes, say: 'Compare your equation for Reaction B with your partner. Count the atoms on each side — are they balanced? [10-second wait time.] If your partner got a different answer, work it out together.' [Allow 2 minutes pair checking.] Cold-call one learner to write Reaction A on the board, and a different learner to write Reaction B. Ask the class: 'Thumbs up if you agree with Reaction B — is the calcium balanced? The chlorine? The carbon? The oxygen? The hydrogen? Check atom by atom.' [10-second wait time between each element check.] After equations are confirmed, say: 'Now here is the big connection — Mwangi, you made mandazi last week for the school market. What makes the dough rise?' [Accept response.] 'Yes — baking soda! And baking soda is $NaHCO_3$ — a hydrogen carbonate. The same type of reaction: acid + hydrogen carbonate gives a salt, water, and CO_2. Write that equation in your Phenomenon Connection box.' After equations, say: 'Look at Reaction A — NaCl is the product. Does anyone recognise NaCl from our phenomenon?' [Cold-call 2 learners, 10-second wait time.] 'Yes — it is table salt, the one that did NOT change overnight. Why might $CaCl_2$ behave differently? We will explore that in Lesson 6. For now, write: $CaCl_2$ is	Counting (NGSS SEP 5 — Using Mathematics and Computational Thinking) — The step-by-step scaffold (word equation → unbalanced symbol equation → balanced equation with state symbols) prevents learners from jumping to an answer without reasoning. The atom-by-atom class verification is a public sensemaking routine that normalises checking and error correction. The Phenomenon Connection box forces an explicit cross-context link: laboratory reaction ↔ mandazi kitchen ↔ anchoring phenomenon (CAN fertiliser and washing soda), deepening understanding that CO_2 production from carbonates/hydrogen carbonates is a universal chemical pattern, not an isolated lab event.	Phase 3. Observable indicators: (1) Reaction A equation is correctly balanced: $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$ with correct state symbols; (2) Reaction B equation is correctly balanced with a coefficient of 2 on HCl: $2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$; (3) Phenomenon Connection box identifies CO_2 as the gas linking the lab reaction to the mandazi phenomenon. If more than 30% of learners omit the coefficient of 2 in Reaction B, teacher re-teaches atom counting at the start of Lesson 5.

		hygroscopic — it absorbs water from the air.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Precipitation reactions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Each group receives a blank A5 card and constructs a 'mini flow chart' for the two preparation methods learned today (acid + base → $NaCl + H_2O$; acid + carbonate → $CaCl_2 + CO_2 + H_2O$), using arrows and colour coding (base = blue, carbonate = green, products = black). Groups post their cards on the DQB in the section labelled 'Experiment 2b Evidence'. The class then does a 'Gallery Walk' (2 minutes) to read other groups' cards. Back at their seats, learners individually update their personal DQB tracker sheet, answering: 'What new evidence from today helps us answer the driving question? What new questions does today's experiment raise?' Representative responses are shared (cold-call 2 learners). The teacher records one new class-level question on the DQB: 'If $CaCl_2$ absorbs water from the air, could we use it to DRY things? Is this connected to why CAN fertiliser clumps?' This question is tagged as 'To be answered in Lesson 6 — Behaviour of Salts in Air.'	Distribute A5 cards. Say: 'You have 4 minutes to draw your mini flow chart for today's two methods. Use the colour code on the board: base reactions in blue, carbonate reactions in green, all products in black. Go.' [Set 4-minute timer.] After 4 minutes: 'Post your card on the DQB — Experiment 2b section. Now stand up and do a 2-minute gallery walk. Read at least two other groups' cards. Look for: is their equation balanced? Is the CO_2 state symbol correct? Sit back down when the timer sounds.' [Set 2-minute timer.] After gallery walk, say: 'Update your personal tracker now. Two questions: What evidence did today give us toward the driving question? What new questions do you have? Two minutes.' [Allow 2 minutes.] Cold-call 2 learners (not the same learners cold-called in Phase 1): 'Akinyi, what new evidence does today give us for the driving question?' [10-second wait time.] 'Otieno, what new question came up for you today?' [10-second wait time.] Record Otieno's or a similar question on the DQB and tag it. Say: 'Notice how our DQB is growing — every lesson, we add more evidence AND more questions. This is exactly how real chemists at KEBS [Kenya Bureau of Standards] and KARLO [Kenya Agricultural and Livestock Research Organisation] work — evidence leads to new questions.'	Gallery Walk + Personal DQB Tracker Update (NGSS SEP 8 — Obtaining, Evaluating, and Communicating Information) — The Gallery Walk creates a low-stakes peer-review routine: learners evaluate each other's flow charts for scientific accuracy (balanced equations, correct state symbols) without formal grading. The personal tracker update is a metacognitive routine — learners explicitly articulate what they now know and what they still wonder, building the habit of distinguishing evidence from inference and current knowledge from open questions. Posting questions on the DQB publicly validates scientific curiosity as a classroom norm.	Teacher reviews 6 mini flow charts posted on the DQB during the gallery walk. Observable indicators: (1) Both preparation methods are represented with correct reactants and products; (2) CO_2 is shown as a product of the carbonate reaction (with 'g' state symbol); (3) At least one group explicitly labels NaCl as 'table salt — unchanged in phenomenon' and/or $CaCl_2$ as 'hygroscopic — linked to phenomenon'. Teacher notes any group that omits CO_2 or water as products for targeted follow-up in Phase 5.
Model Building Phase 📹 **VIDEO:**	Learners retrieve their cumulative 'Soluble Salt Preparation Methods' flow chart begun in Lesson 2 and extended in Lessons 2 and 3.	Say: 'Get out your cumulative flow chart — the one we have been building since Lesson 2. Today you complete the final two	Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models) — The ongoing flow chart is a visible, tangible	Teacher circulates during model building and uses a 3-point checklist (on clipboard): (1) Learner has added both new

Precipitation reactions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Today they add the final two branches: (4) Acid + Base → Salt + Water (example: $HCl + NaOH \rightarrow NaCl + H_2O$) and (5) Acid + Carbonate/Hydrogen Carbonate → Salt + CO_2 + Water (example: $HCl + CaCO_3 \rightarrow CaCl_2 + CO_2 + H_2O$; and $HCl + NaHCO_3 \rightarrow NaCl + CO_2 + H_2O$). Learners annotate their flow chart with a 'Phenomenon Link' arrow: from the 'CO_2 produced' branch, an arrow points to a sticky note reading 'Same gas as in mandazi baking! Same type of reaction as washing soda + moisture?' They also annotate the NaCl branch: 'This is the table salt that did NOT change in the storeroom — why? (Revisit in Lesson 6).' Learners update the chart title to: 'All Soluble Salt Preparation Methods — Evidence gathered in Lessons 1–4.' The completed chart will be submitted at the end of Lesson 8 as a summative evidence portfolio artefact.	branches. You have 6 minutes. Use today's balanced equations directly from your notebook. Add your phenomenon link annotations.' [Set 6-minute timer. Circulate and prompt: 'Does your CO_2 branch link back to our mandazi observation? Show me where on your chart.' Use targeted questioning for 2–3 learners who struggled in Phase 3: 'What are the products of acid + carbonate? Check your notebook — write them on the branch.'] After 6 minutes, say: 'Hold up your flow chart. Look at it. You now have the complete picture of how chemists prepare soluble salts — five methods, all connected to real Kenyan chemistry: our CAN fertiliser, our washing soda, our table salt, our mandazi. This chart is your growing model of salts.' [Pause for effect — 5 seconds.] Close with: 'In Lesson 5 we will tackle insoluble salts — prepared by a completely different method called precipitation. You already predicted it might involve mixing two solutions. Keep that prediction in mind.' Collect any learner models that need teacher annotation feedback (select 3–4 voluntarily) and return at the start of Lesson 5.	representation of growing scientific understanding. Each lesson's additions are colour-coded by date, so learners can literally see their knowledge building over time. Annotating the chart with phenomenon links (mandazi CO_2, table salt unchanged, CAN clumping) embeds the NGSS crosscutting concept of Patterns — learners recognise that the same gas (CO_2) appears in cooking, in industry, and in the atmospheric behaviour of salts, and that these are not coincidences but expressions of the same underlying chemistry.	branches with correct reactants and products; (2) Learner has included balanced equations (not just word equations) on the new branches; (3) Learner has added at least one phenomenon link annotation. Teacher records checklist data per learner name for Lessons 1–4 cumulative tracking. Any learner scoring 1/3 receives a brief targeted question: 'Show me where CO_2 appears on your chart — and where does CO_2 also appear in our driving question phenomenon?' to prompt revision before Lesson 5.

D. TEACHER REFLECTION

After this lesson, reflect on the following: (1) EQUATION BALANCING: Did the majority of learners correctly write the coefficient of 2 in $2HCl + CaCO_3 \rightarrow CaCl_2 + CO_2 + H_2O$ without prompting? If not, plan a 5-minute 'atom counting warm-up' at the start of Lesson 5 using a simple example (e.g., $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$) before introducing precipitation. (2) CO_2 GAS TEST: Did all groups successfully observe milky limewater? If any groups failed, identify whether the cause was equipment (loose bung, delivery tube not submerged) or procedural (too little HCl added). Adjust equipment setup instructions for Lesson 5 if needed. (3) PHENOMENON CONNECTION: During Phase 3, how many learners spontaneously made the mandazi–CO_2–washing soda connection without being prompted? If fewer than half the class made this connection independently, the phenomenon link is not yet internalised — plan an explicit 'phenomenon revisit' moment in Lesson 5 before introducing precipitation. (4) SAFETY COMPLIANCE: Were all PPE protocols followed throughout? Were waste solutions neutralised before disposal? Note any lapses for a class safety briefing at the start of Lesson 5. (5) DQB QUALITY: Did learners generate genuine new questions on the DQB tracker, or did they simply restate today's lesson

content? High-quality questions (e.g., 'Can $CaCl_2$ be used as a drying agent in Kenyan food storage?') indicate conceptual transfer. Record the most generative student questions and use them as lesson openers in Lessons 5 and 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed: (1) Universal indicator changing from red/orange to green during neutralisation of HCl with NaOH, indicating pH ≈ 7 at the endpoint; (2) Vigorous fizzing/effervescence when HCl was added to marble chips ($CaCO_3$); (3) Limewater turning milky/cloudy when CO_2 gas was bubbled through it — confirming the identity of the gas produced in the acid + carbonate reaction; (4) No visible solid precipitate formed in either reaction (both products were soluble salts in solution).
What did I learn?	Learners learned: (1) Soluble salts can be prepared by reacting an acid with a base (neutralisation) — no gas is produced, and the endpoint is identified using an indicator (pH ≈ 7); (2) Soluble salts can also be prepared by reacting an acid with a carbonate or hydrogen carbonate — CO_2 gas and water are also produced alongside the salt; (3) The balanced equation for neutralisation is: $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$; (4) The balanced equation for acid + carbonate is: $2HCl(aq) + CaCO_3(s) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$; (5) The same type of reaction (acid + hydrogen carbonate → salt + CO_2 + water) explains why mandazi dough rises when baking soda is added; (6) NaCl (table salt) and $CaCl_2$ (calcium chloride) are both soluble salts but may behave differently when exposed to air — this will be investigated in Lesson 6.
How does this explain the phenomenon?	Learners can now explain: (1) Why there is no visible change during neutralisation even though a chemical reaction is occurring — the salt (NaCl) and water produced are both colourless and soluble, so an indicator is needed to confirm the reaction is complete; (2) Why fizzing occurs when HCl reacts with $CaCO_3$ — CO_2 gas is produced as a product of the acid + carbonate reaction, and this same gas is responsible for the rising of mandazi dough; (3) How to write and balance chemical equations for both preparation routes, including correct state symbols; (4) How today's evidence connects to the driving question — NaCl (prepared by neutralisation) is the table salt that did not change overnight, while $CaCl_2$ (prepared by acid + carbonate) is hygroscopic and may explain the clumping behaviour observed in CAN fertiliser, to be confirmed in Lesson 6.

LESSON 5 (80 minutes): Preparing Insoluble Salts — Double Decomposition and Precipitation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To prepare an insoluble salt by double decomposition (precipitation) and compare this method to direct synthesis, linking preparation method choice to salt solubility.
Knowledge	Learners explain that insoluble salts cannot be prepared by dissolving methods and must be prepared by mixing two aqueous solutions of soluble salts (double decomposition/precipitation); they write full and ionic equations for precipitation reactions (e.g. $BaSO_4$ and PbI_2); they state the solubility rules for sulphates, chlorides, carbonates, and nitrates.
Skills	Learners carry out a precipitation experiment safely; filter and wash a precipitate; write balanced molecular and ionic equations for double decomposition reactions; compare preparation methods and select the correct method for a given salt based on its solubility.
Attitudes	Learners appreciate the importance of choosing the correct preparation method for a given salt and develop responsibility in safe handling and proper disposal of chemical waste (including barium compounds, which are toxic).
Key Inquiry Question	How can salts be prepared in the laboratory?
Purpose in Storyline	In previous lessons learners prepared soluble salts (by acid–metal, acid–base, and acid–carbonate reactions). But barium sulphate — an insoluble salt — cannot be made that way. This lesson asks: if CAN fertiliser's ammonium nitrate is soluble and can be made by neutralisation, why can't we make barium sulphate the same way? Learners discover that the solubility of the target salt dictates the preparation method, giving them another piece of the puzzle: understanding why different salts (like those in the phenomenon — CAN, washing soda, table salt) have different properties begins with knowing how and why they are made differently.
Safety Notes	Barium chloride ($BaCl_2$) is TOXIC — learners must wear gloves and safety goggles at all times when handling it; no mouth pipetting; wash hands thoroughly after the experiment. If lead(II) nitrate / lead(II) iodide is used as the alternative, treat all lead compounds as toxic and carcinogenic — do not use with younger or vulnerable learners without full risk assessment. All filtrate and washings containing barium or lead ions must be collected in a labelled waste beaker and handed to the teacher for safe disposal — do NOT pour down the sink. Broken glassware must be reported immediately. PCIs (Environmental Issues / Safety in the Laboratory) apply throughout this lesson.

B. LESSON OVERVIEW

Learners build on their knowledge of solubility rules (Lesson 3) and soluble-salt preparation methods (Lessons 3–4) to investigate a fundamentally different preparation route: double decomposition (precipitation). Working in groups, they mix solutions of barium chloride and sodium sulphate (or, as an alternative safer option, lead(II) nitrate and potassium iodide) and immediately observe a precipitate of an insoluble salt forming. They filter, wash, and dry the product, then write full balanced equations and net ionic equations. A class comparison chart consolidates all four preparation methods studied so far. The Driving Question Board is updated to address method selection, and the cumulative model is revised to show that the choice of preparation method is governed by the solubility of the target salt — directly connecting back to why CAN (soluble, made by neutralisation) and barium sulphate (insoluble, made by precipitation) are chemically different entities.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** Meaningfully composing functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Reaction Overview (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown two labelled beakers of clear, colourless solutions: Beaker A (barium chloride solution, 0.1 mol/L) and Beaker B (sodium sulphate solution, 0.1 mol/L). The teacher states: 'Both of these solutions contain salts that are fully dissolved in water. I am going to pour Beaker A into Beaker B. In your exercise books, draw what you predict you will see — and write one sentence explaining your prediction using what you already know about solubility rules from Lesson 3.' Learners write individually for 3 minutes, then share predictions with a shoulder partner. The teacher cold-calls 4 learners (two who predict a precipitate, two who predict no change) to read their predictions aloud, recording key words — 'precipitate', 'stays clear', 'white solid', 'insoluble' — on the board without confirming or denying. The teacher connects to the phenomenon: 'Last night in our storeroom, CAN fertiliser pulled in moisture from the air. Today we are asking a different question: what happens when two dissolved salts meet each other? Could this ever produce something like the white crust we saw on the washing soda?'	1. Display the two beakers prominently; point to each label and ask: 'What dissolved salt is in each beaker?' (Wait 10 seconds; cold-call 2 learners.) 2. Pose the prediction task verbally and in writing on the board: 'What will you observe when these two solutions are mixed? Use your solubility rules.' 3. Allow 3 minutes of silent individual writing. 4. Pair-share for 1 minute. 5. Cold-call 4 learners; scribe their key prediction words on the board under two columns: 'Predicts precipitate' / 'Predicts no change'. 6. Say: 'We will come back to these predictions the moment we do the experiment. Do NOT rub them out.' 7. Bridge to phenomenon: 'The white powdery crust on the washing soda — was that a precipitation? Let's keep that question open as we work today.'	Predict–Record–Revisit: Learners commit predictions to writing before observing, activating prior knowledge of solubility rules. Recording predictions publicly (on the board) creates cognitive accountability — learners are motivated to reconcile their prediction with the evidence they will gather. This strategy surfaces alternative conceptions (e.g. 'all salt solutions mixed together stay clear') that the teacher can address during the Explain phase.	Observable indicator: Every learner has a written prediction in their exercise book that references a solubility rule (e.g. 'barium sulphate is insoluble, so a solid will form'). The teacher circulates during the 3-minute writing time and scans for predictions that do NOT reference solubility — these learners are flagged for targeted questioning during the Explain phase. Cold-call responses reveal whether learners can correctly recall that $BaSO_4$ is one of the insoluble sulphate exceptions.
Observe Phase 📹 **VIDEO:** Meaningfully composing functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups of 4 carry out the precipitation experiment following a teacher-guided procedure displayed on the board. PRIMARY EXPERIMENT (Barium sulphate): Each group measures 10 cm^3 of 0.1 mol/L barium chloride solution into a boiling tube, then adds 10 cm^3 of 0.1 mol/L sodium sulphate solution dropwise while swirling. Learners observe the immediate	1. Before starting: conduct a safety briefing — 'Barium chloride is toxic. Gloves and goggles on before you touch the solutions. Show me your gloves.' (Visually confirm all groups before proceeding.) 2. Demonstrate the first 3 steps at the front bench: measuring, adding dropwise, swirling. Say: 'Watch the moment the first drop of sodium sulphate	Structured Observation Table with Mid-Experiment Pause: The observation table scaffolds learners to record discrete, comparable data points (before/during/after mixing) rather than narrative prose, enabling direct comparison across groups. The mid-experiment pause — where learners connect the instant precipitate to the slower	Observable indicators: (1) All learners record all four columns of the observation table with specific, accurate descriptions (e.g. 'dense white precipitate forms immediately on mixing' — not vague terms like 'something happened'). (2) At least one group member in each group can explain why the precipitate forms by referencing solubility ('barium

📖 **HTML:** Chemical Reaction Overview (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	formation of a dense white precipitate of barium sulphate. They filter the mixture using filter paper and funnel into a conical flask, wash the residue on the filter paper twice with distilled water, and transfer the filter paper + precipitate to a watch glass to dry. ALTERNATIVE EXPERIMENT (Lead(II) iodide — if barium chloride is unavailable): Mix 10 cm^3 of 0.1 mol/L lead(II) nitrate solution with 10 cm^3 of 0.1 mol/L potassium iodide solution — a vivid golden-yellow precipitate of PbI_2 forms immediately, which is visually dramatic and highly motivating. Learners record observations in a structured table: Observation Before Mixing / Observation on Mixing / Colour of Precipitate / Filtrate Colour After Filtering. They also attempt to write the word equation and symbol equation in their books during the filtering wait time. The teacher connects to the phenomenon mid-experiment: 'You are making a new insoluble salt right now in your test tube — just as nature made the white crust on the washing soda by a different process. But notice: your precipitate forms instantly when two solutions meet. The white crust on the washing soda took overnight. What does that tell us about the two processes?'	enters — tell me exactly what you see.' (Wait 15 seconds of silent watching during demo.) 3. Release groups to begin; circulate and prompt: 'Describe the precipitate — colour? texture? does it settle or float?' 4. At the filtering stage, pose a thinking question while learners wait: 'Why do we wash the precipitate with distilled water rather than tap water?' (Wait 10 seconds; cold-call 3 groups for answers.) 5. Mid-experiment phenomenon connection — state the quoted bridge question above, then give 2 minutes for groups to discuss and jot a brief answer. 6. At end: collect all filtrate and washings in a labelled waste beaker. Say: 'Pour nothing down the sink — barium ions are toxic to aquatic life in Lake Victoria and our local rivers.' 7. Cold-call 2 groups to read their observation tables aloud before moving to Explain.	efflorescence of washing soda — is a Bridging Analogy strategy: it uses the familiar phenomenon to build conceptual contrast between precipitation (ionic recombination in solution) and efflorescence (loss of water of crystallisation to air), preventing learners from conflating the two processes.	sulphate is insoluble in water'). (3) Learners correctly identify and separate waste for disposal — teacher checks every group's waste beaker before they leave the practical area. Flag: any group that cannot distinguish between the residue (precipitate = product) and the filtrate (waste) needs re-teaching before Explain phase.
Explain Phase 📹 **VIDEO:** Meaningfully composing functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The class reconvenes. The teacher reveals the correct predictions on the board and invites learners to evaluate their own predictions. Learners then work through three layers of explanation, guided by the teacher: LAYER 1 — Word and symbol equations. Groups write	1. Return to the prediction board: 'Hands up if your prediction was correct. Now — more importantly — can you explain WHY it was correct or incorrect using today's chemistry?' (Wait 10 seconds; cold-call 3 learners — deliberately include one whose prediction was wrong.) 2. Layer 1 — whiteboards:	Three-Layer Equation Scaffolding (Word → Molecular → Net Ionic): This progressive scaffolding strategy moves learners from familiar language (word equation) through symbol representation (balanced molecular equation) to the most powerful chemical model (net ionic equation), each layer	Observable indicators: (1) Learners write a correctly balanced molecular equation with state symbols (aq) and (s) correctly assigned. (2) Learners correctly identify Na^+ and Cl^- as spectator ions and write a correct net ionic equation. (3) The Method Comparison Chart column is

📖 **HTML:** [Chemical Reaction Overview (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	the balanced molecular equation on mini-whiteboards or in books: $BaCl_2(aq) + Na_2SO_4(aq) \rightarrow BaSO_4(s) + 2NaCl(aq)$. The teacher cold-calls two groups to share and the class checks balancing together. LAYER 2 — Ionic equations. The teacher introduces the concept of spectator ions using a Think-Aloud: 'If $BaCl_2$ is dissolved, it exists as Ba^{2+} and $2Cl^-$ ions. Na_2SO_4 exists as $2Na^+$ and SO_4^{2-} ions. Which ions actually come together to form the precipitate?' Learners write the full ionic equation, then cancel spectator ions to produce the net ionic equation: $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$. Learners annotate: 'The Cl^- and Na^+ ions are spectators — they do nothing; they just stay in solution.' LAYER 3 — Method Comparison Chart. The teacher reveals a four-column comparison chart (started in Lesson 3) on the board. Groups fill in the 'Precipitation / Double Decomposition' column: target salt type = insoluble; reactants = two soluble salt solutions; driving force = formation of insoluble product. Learners then answer the key question in their books: 'Why can't we make barium sulphate by reacting barium metal with sulphuric acid directly in the lab?' (Answer: barium is a reactive Group 2 metal — the reaction would be violently exothermic and dangerous; and $BaSO_4$ is insoluble so it would coat the metal and stop the reaction.) Phenomenon connection: 'So: CAN fertiliser is ammonium nitrate — it IS soluble, and it IS made industrially by neutralisation. Barium sulphate is insoluble and MUST be made by	'Write the balanced molecular equation for the reaction. You have 2 minutes. Go.' Circulate; look for incorrect balancing of NaCl (coefficient of 2). Cold-call one correct and one incorrect group to compare. 3. Layer 2 — Think-Aloud: slow, deliberate, write every ion on the board as you speak. Say: 'I am going to split every soluble compound into its ions — watch carefully. Tell me: which ions appear on BOTH sides of the equation unchanged?' (Wait 15 seconds; cold-call 4 learners.) Say: 'Those are our spectator ions — cross them out. What remains is the NET ionic equation. Write it.' 4. Layer 3 — chart: display the partially completed chart; give groups 3 minutes to complete their column; cold-call groups to contribute one cell each. 5. Pose the 'why not barium + sulphuric acid?' challenge question — wait 15 seconds — cold-call 2 learners; confirm and extend. 6. Deliver the phenomenon bridge statement (quoted above) deliberately and slowly; ask learners to write the key idea in their own words in one sentence.	building on the previous. The Method Comparison Chart is a Concept Mapping / Anchor Chart strategy: it makes the organising principle — solubility determines preparation method — visible and cumulative across all five preparation methods, preventing isolated memorisation and building a coherent conceptual framework. The 'why not direct synthesis?' challenge question uses Cognitive Conflict to deepen understanding of method selection.	correctly and completely filled in. (4) Every learner has written a one-sentence summary connecting solubility to method choice. Teacher exit check: ask 3 randomly selected learners to state one insoluble salt and which method would be used to prepare it — if fewer than 2/3 answer correctly, the teacher returns to the comparison chart before proceeding to the DQB.

	precipitation. The solubility rule determines the method. This is why knowing solubility rules is not just abstract chemistry — it determines how every salt in agriculture, medicine, and industry is manufactured.'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Meaningfully composing functions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Chemical Reaction Overview (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes of different colours. On the FIRST sticky note (green = 'we can now answer this'), each group writes an answer to the standing DQB question: 'How can salts be prepared in the laboratory?' — specifically adding the precipitation method to their answer. On the SECOND sticky note (orange = 'new question this lesson raised'), groups write a new question that arose today. Sample new questions anticipated: 'How do we know which method to choose without testing?', 'Are there other insoluble salts made this way in industry?', 'Why is barium sulphate used in X-rays if it is toxic?', 'Is the white crust on washing soda also a kind of precipitation?'. Groups stick both notes on the class DQB under the appropriate columns. The teacher reads 3–4 orange notes aloud and says: 'These are exactly the questions a chemist would ask. The question about barium sulphate and X-rays — that is applications of salts, which we will reach. The question about washing soda's white crust — is that precipitation? Let's hold that — does anyone want to offer a quick thought?' (Brief 2-minute open discussion, no resolution yet.) The teacher writes the new organising question on the DQB header: 'How do we choose the right preparation method for a given salt?' and draws an arrow	1. Distribute sticky notes; state the task clearly: 'Green = what we can now answer. Orange = a new question this lesson raised. You have 4 minutes.' 2. Circulate; prompt groups who are stuck on the orange note: 'What surprised you today? What do you still not understand?' 3. Invite groups to post notes; read 3–4 orange notes aloud with genuine curiosity. 4. Pose the washing soda mini-discussion question (quoted above); enforce 2-minute time limit; do not resolve — say: 'We will return to this when we study salt behaviour in air.' 5. Write and underline the new DQB organising question; draw the arrow to the solubility rules card. 6. Say: 'Look at our DQB. Each lesson we have added a new preparation method AND a new question. This is exactly how real science works — every answer opens new questions.' 7. Photograph the DQB for the ARES platform record.	Two-Colour Sticky Note DQB Protocol: Using two distinct colours for 'answered' vs 'new questions' makes the growth of understanding visually explicit — learners can see the board filling with green (answers) and orange (new questions) over the sequence of lessons. This is a Metacognitive Monitoring strategy: it trains learners to distinguish between what they know and what they do not yet know, a core scientific practice. Connecting the new organising question to the existing solubility rules card with a drawn arrow models Systems Thinking — showing learners that chemistry concepts are interconnected, not isolated facts.	Observable indicators: (1) Every group produces a green sticky note that correctly names precipitation/double decomposition as a preparation method for insoluble salts and gives an example. (2) Every group produces an orange sticky note with a genuine, chemistry-relevant question (not a repeat of a question already answered). (3) The teacher checks that no green sticky note contains a misconception (e.g. 'precipitation is used for all salts') — any such note is addressed publicly by asking the class to refine it before it is posted.

	connecting it to the solubility rules card posted in Lesson 3.			
Model Building Phase 📹 **VIDEO:** Meaningfully composing functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Chemical Reaction Overview (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative 'Salt Preparation Decision Tree' model started in Lesson 3 and updated in Lesson 4. Today they add a new branch: 'Is the target salt INSOLUBLE?' → YES → Use Double Decomposition (Precipitation): mix two soluble salt solutions, one containing the target cation and one containing the target anion → filter, wash, dry precipitate. They annotate the branch with: the net ionic equation for $BaSO_4$ or PbI_2, the names of spectator ions, and a note: 'Spectator ions remain in filtrate — collect as waste, dispose safely.' Learners also add a 'Kenyan context' annotation to the insoluble salts branch: 'Barium sulphate ($BaSO_4$) is used in barium meals for stomach X-rays at Kenyatta National Hospital — it is safe to swallow because it is INSOLUBLE and cannot be absorbed into the bloodstream, despite barium ions being toxic.' Finally, learners draw a two-way comparison arrow between the new branch (insoluble salts — precipitation) and the existing branches (soluble salts — neutralisation, acid-metal, acid-carbonate), labelling it: 'SOLUBILITY of target salt determines method choice.' The teacher poses the closing model prompt: 'Using your decision tree, predict which method you would use to prepare calcium carbonate — a salt found in coral reefs off the Kenyan coast at Malindi. Be ready to justify your choice next lesson.'	1. Say: 'Take out your decision tree model. We are adding a new branch today — the most important one yet, because it breaks the pattern of all our previous methods.' (Pause 5 seconds to build anticipation.) 2. Guide the model update step by step — draw each element on the board as learners add it to their own models. 3. Introduce the KNH barium meal context: 'Has anyone heard of a barium meal at a hospital? Why would doctors give a patient a barium solution to drink if barium ions are toxic?' (Wait 15 seconds; cold-call 3 learners.) Confirm: 'Because $BaSO_4$ is INSOLUBLE — it cannot dissolve into the blood. The same property that makes it hard to prepare in the lab makes it safe in medicine.' 4. Instruct learners to add the comparison arrow and label it — circulate to check labelling accuracy. 5. Pose the calcium carbonate closing prompt (quoted above) — give 2 minutes for individual written responses in exercise books. Cold-call 2 learners to share. Do not confirm the answer — say: 'Hold your prediction. Lesson 6 will tell you if you were right.' 6. Close: 'Look at your driving question on the board. We now have answers for four of the five preparation methods. One more lesson of preparation, then we turn to the big question — why does CAN get wet, washing soda shrinks, and table salt does nothing?'	Cumulative Decision Tree Model Revision (NGSS Practice 2 — Developing and Using Models): The decision tree serves as a running scientific model that learners physically update each lesson, making their growing understanding tangible and structured. Adding the 'Kenyan context' annotation (barium meal at KNH) is a Real-World Anchoring strategy: it connects abstract ionic chemistry to a biomedical application familiar from Kenyan healthcare, reinforcing that solubility is not merely an exam concept but a property with life-or-death implications. The closing 'calcium carbonate' prediction task is a Transfer and Application strategy that checks whether learners can use the model as a predictive tool — the hallmark of genuine understanding.	Observable indicators: (1) Every learner's decision tree has a correctly labelled new branch for precipitation/double decomposition with the net ionic equation, spectator ion annotation, and safe disposal note. (2) The comparison arrow is drawn and labelled correctly ('solubility of target salt determines method choice'). (3) The closing calcium carbonate prediction is written in the exercise book with a justification referencing solubility rules ($CaCO_3$ is insoluble → use precipitation). Teacher collects exercise books at the end of the lesson to review the calcium carbonate prediction and decision tree update — these serve as the primary written formative assessment record for this lesson.

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) Did all learners correctly distinguish between spectator ions and reacting ions when writing net ionic equations — or did some learners cancel ALL ions, leaving a blank equation? If so, re-teach the concept that the precipitate's ions are REACTING ions, not spectators, using a physical analogy (e.g. two people shaking hands = reacting ions; bystanders watching = spectator ions). (2) Were learners able to connect solubility rules to method choice independently, or did they rely entirely on the teacher's prompt? If method selection is not yet internalised, consider a short card-sorting activity at the start of Lesson 6: give groups cards with salt names and ask them to sort into 'soluble — neutralisation method' vs 'insoluble — precipitation method' before any teaching. (3) Safety: were all groups' barium/lead waste correctly collected and labelled? If any filtrate was poured down the sink, address this directly at the start of Lesson 6 as a PCI (Environmental Issues) discussion moment, connecting unsafe disposal to contamination of local water sources. (4) Did the DQB update generate genuine new questions, or were learners producing formulaic responses? If the latter, try modelling a 'good orange sticky note' at the start of Lesson 6 using a think-aloud. (5) Did the KNH barium meal context engage learners meaningfully — did it produce visible curiosity or discussion? Note which groups responded most — this may indicate learners with personal or family experience of hospital investigations, which could be built into future lessons on applications of salts.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	When barium chloride solution and sodium sulphate solution were mixed, a dense white precipitate formed immediately. The precipitate was filtered, washed with distilled water, and collected on filter paper. The filtrate (waste solution) remained clear. An alternative experiment using lead(II) nitrate and potassium iodide produced a vivid golden-yellow precipitate of lead(II) iodide.
What did I learn?	Insoluble salts cannot be prepared by the methods used for soluble salts (acid–metal, acid–base, acid–carbonate). They are prepared by double decomposition (precipitation): two soluble salt solutions, each supplying one ion of the target salt, are mixed; the insoluble salt precipitates immediately. The net ionic equation shows only the reacting ions (e.g. $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$); the remaining ions ($Na^+$ and Cl^-) are spectator ions. The solubility of the target salt determines which preparation method must be used.
How does this explain the phenomenon?	The anchoring phenomenon involves three salts behaving differently in air: CAN (ammonium nitrate — soluble, made by neutralisation), washing soda (sodium carbonate decahydrate — soluble, efflorescent), and table salt (sodium chloride — soluble, stable). Today's lesson showed that not all salts are soluble — some, like barium sulphate, are insoluble and must be made by a completely different method. This deepens our understanding of the phenomenon: the properties of a salt (solubility, stability in air, reactivity) are linked to its composition and the method by which it was formed. Choosing the right preparation method — and understanding why — is essential for producing salts safely and correctly for use in agriculture (fertilisers), medicine (barium meals at KNH), and industry across Kenya.

LESSON 6 (80 minutes): Behaviour of Salts in Air: Hygroscopic, Deliquescent and Efflorescent Salts	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners investigate how different salts behave when exposed to air, gathering direct experimental evidence to fully explain the anchoring phenomenon (CAN fertiliser clumping, washing soda crusting, table salt unchanged) and provide the first complete answer to the unit driving question.
Knowledge	Learners will be able to: (c) describe the behaviour of salts when exposed to air, distinguishing between hygroscopic, deliquescent, and efflorescent salts using experimental evidence and mass-change data.
Skills	Learners will be able to: (b) carry out a controlled experiment exposing anhydrous calcium chloride, sodium carbonate decahydrate, and sodium chloride to air, measuring and recording mass changes and visual observations accurately; write balanced equations for efflorescence and deliquescence where applicable.
Attitudes	Learners will be able to: (e) appreciate the applications of salts in day-to-day life, recognising why understanding salt behaviour in air is essential for safe and effective storage of fertilisers, food preservatives, and laboratory chemicals in Kenya.
Key Inquiry Question	How do salts behave when exposed to air?
Purpose in Storyline	This is Lesson 6 of 8 in the evidence-gathering sequence. By the end of this lesson, learners will hold enough experimental evidence to fully explain why CAN fertiliser granules clumped (deliquescence), why washing soda crystals shrank and formed a white crust (efflorescence), and why table salt remained unchanged — directly answering the unit driving question for the first time and updating the class Driving Question Board with a complete, evidence-backed explanation.
Safety Notes	Anhydrous calcium chloride is a skin and eye irritant — learners must wear safety goggles and avoid touching their faces. Sodium carbonate decahydrate may cause mild skin irritation; wash hands after handling. All chemicals must be handled in small quantities (approx. 2 g per sample). Dispose of all salt samples in the designated waste container — do NOT pour down the sink. Learners with known respiratory conditions should sit away from open chemical containers. The teacher must inspect all watch glasses/petri dishes before distribution.

B. LESSON OVERVIEW
Learners begin by revisiting the anchoring phenomenon and the DQB questions gathered across Lessons 1–5. They then carry out Experiment 3: three labelled watch glasses, each holding a pre-weighed sample of anhydrous calcium chloride ($CaCl_2$), sodium carbonate decahydrate ($Na_2CO_3 \cdot 10H_2O$), and sodium chloride ($NaCl$), are left exposed to classroom air for 20–25 minutes while learners predict and record observations at timed intervals. Mass changes are recorded using a digital balance. Learners use their data to define hygroscopic, deliquescent, and efflorescent behaviour, then map each salt directly onto the anchoring phenomenon. The lesson closes with the DQB receiving its first complete written answer to the driving question, and learners construct or revise their cumulative particle-level model to show water molecule interactions at a salt surface.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**

Predict Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown the three pre-weighed watch glasses (labelled A: $CaCl_2$, B: $Na_2CO_3 \cdot 10H_2O$, C: NaCl) before exposure begins. Working individually, each learner writes a prediction in their science notebook: 'After 25 minutes in open air, which sample will gain mass, lose mass, or stay the same — and why?' They then share predictions with their table partner (Think-Pair) and each pair records one joint prediction on a sticky note posted under the anchoring phenomenon photograph on the DQB. The teacher cold-calls 4 pairs to read their predictions aloud before the experiment starts, so predictions are publicly committed.	Display the three labelled watch glasses on the front bench and show learners the digital balance reading for each (record on the board: A = 2.0 g, B = 2.0 g, C = 2.0 g). Say: 'You have studied salts for five lessons. Before we collect any new data today, I want you to commit to a prediction — in writing, individually, no group discussion yet.' Allow 3 minutes of silent individual writing. Then say: 'Now turn to your partner. You have 2 minutes. Agree on ONE joint prediction for each sample and write it on your sticky note.' After 2 minutes, cold-call 4 pairs (aim for at least one pair predicting mass gain, one predicting mass loss, one predicting no change) to read predictions aloud. Post sticky notes on the DQB under the label 'Our Predictions — Lesson 6'. Do NOT confirm or correct predictions. Say: 'We will return to every one of these predictions with data. Let the experiment decide.'	Think-Pair-Share with public commitment: By posting written predictions on the DQB before the experiment, learners create cognitive accountability — they must later reconcile their prediction with evidence, driving deeper conceptual revision rather than passive reception of results.	Teacher scans sticky notes during posting. Observable indicator: Does the learner's prediction reference a chemical or physical reason (e.g., '$CaCl_2$ will gain mass because it attracts water from the air') or is it purely a guess? Learners who give purely intuitive predictions (no reasoning) are flagged for additional prompting during the observe phase. Teacher notes at least 3 contrasting predictions on the board to create productive disagreement.
Observe Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4 carry out Experiment 3 simultaneously. Each group receives three pre-weighed watch glasses (A: ~2 g anhydrous $CaCl_2$, B: ~2 g $Na_2CO_3 \cdot 10H_2O$, C: ~2 g NaCl) and places them in an open area of their bench. Learners record the initial mass of each, then observe and record visual observations and mass readings at 0 min, 10 min, and 20–25 min intervals in a structured results table (provided on the learner worksheet). Observations include: texture, appearance (crystal shape, surface changes, moisture), and any visible powder or liquid formation. At the end of the exposure period, groups weigh each sample again and calculate	Distribute materials. Say: 'Before you place your watch glasses on the bench, record the exact mass of each sample — read the balance to two decimal places. This is your Time 0 reading. Write it in your table NOW.' Circulate; check that all groups have recorded Time 0 before samples are placed on the bench. Set a visible timer on the board for 10 minutes. While samples are exposed, do NOT tell learners what will happen. At the 10-minute mark, say: 'Record your observations for each sample — use your SENSES (sight, careful touch with a glass rod). Record mass again.' At 20–25 minutes, say: 'Final mass reading. Calculate	Structured Data Collection + Class Aggregation Table: Pooling data from all groups onto one class table makes variation visible and builds statistical thinking — learners see that the pattern ($CaCl_2$ gains mass, $Na_2CO_3 \cdot 10H_2O$ loses mass, NaCl unchanged) holds across all groups despite small measurement differences, strengthening the evidence for a real chemical phenomenon rather than experimental error.	Observable indicators: (1) All learners have a completed results table with Time 0, Time 10, and Time 20–25 readings plus a calculated Δm — teacher checks tables during circulation. (2) When cold-called, can the learner correctly identify which sample gained mass and which lost mass? (3) Can the learner correctly match Sample A ($CaCl_2$) to the CAN fertiliser phenomenon without prompting? Groups who cannot make this connection are directed to re-examine the DQB photograph and their data side-by-side.

	mass change (Δm = final − initial). Groups record their data in a shared class data table drawn on the board, allowing comparison across groups. Learners also observe a short teacher demonstration: a small amount of anhydrous $CaCl_2$ in a test tube is exposed to breath moisture — learners observe the immediate surface reaction. Connection to phenomenon: Teacher says, 'Look at sample A. Now look at the photograph of the CAN fertiliser bag on the DQB. What do you notice?' Learners annotate the photograph with evidence arrows.	Δm for each sample. Write your group's values in the class data table on the board.' After all groups have posted data, pause. Give 10 seconds of WAIT TIME, then cold-call 3 groups: 'Group [name], what pattern do you see when you look at ALL groups' data for Sample A?' Repeat for B and C. Then direct attention to the DQB photograph: 'Point to the sample on your bench that behaves most like the CAN fertiliser. Tell your partner why — you have 60 seconds.' Cold-call 2 pairs. Conduct the $CaCl_2$ + breath demonstration and ask: 'What does this tell you about what happens when a salt is exposed to humid air?' WAIT TIME 12 seconds before cold-calling.		
Explain Phase 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Using their experimental data and the class aggregation table, learners construct scientific explanations for each sample's behaviour. The teacher introduces the three terms — hygroscopic, deliquescent, efflorescent — using a Frayer Model graphic organiser (definition, characteristics, examples, non-examples) that learners complete collaboratively in pairs. Learners then write individual scientific explanations (minimum 3 sentences each) for: (a) why $CaCl_2$ gained mass (deliquescent/hygroscopic — absorbs water vapour from air; if enough water is absorbed, it dissolves in the water it absorbs), (b) why $Na_2CO_3 \cdot 10H_2O$ lost mass (efflorescent — loses water of crystallisation to the air, forming anhydrous/partially hydrated powder: $Na_2CO_3 \cdot 10H_2O \rightarrow Na_2CO_3 + 10H_2O$), and (c) why NaCl showed no significant mass change (neither hygroscopic nor	Distribute the Frayer Model organiser. Say: 'I am going to give you three new vocabulary words. Your job — with your partner — is to build the meaning of each word using your experimental data. Do NOT copy from a textbook. Use your OWN data as your evidence.' Allow 8 minutes for pairs to complete Frayer Models. Circulate; if a pair cannot define 'deliquescent', prompt: 'Which sample dissolved in the water it absorbed? What does that tell you about how strongly it attracts water?' After 8 minutes, cold-call 3 pairs to share their definition of one term each. Write the class-agreed definitions on the board. Then say: 'Now write your own explanation — alone, no partner — for why each sample behaved the way it did. Use the word because and refer to your data.' Allow 6 minutes. Cold-call 4 individual learners to read their explanation for one sample. After each	Frayer Model Vocabulary Building + Evidence-Anchored Explanation Writing: The Frayer Model forces learners to build definitions from their own experimental evidence rather than rote memorisation, while the individual written explanation (using 'because' + data reference) requires each learner to make the causal link explicit — a hallmark of scientific reasoning and NGSS Practice 6 (Constructing Explanations).	Observable indicators: (1) Frayer Model contains a learner-generated example from the experiment (not just a textbook example). (2) Individual written explanation explicitly references a data value (e.g., '$CaCl_2$ gained 0.4 g, which shows it absorbed water from the air'). (3) Balanced efflorescence equation is correctly written with correct coefficients. (4) The phenomenon-matching sentences correctly and specifically link all three salts to the three anchoring phenomenon observations. Teacher collects 6 written explanations (one per table group) to review before Lesson 7.

	efflorescent under normal humidity). Learners write the balanced equation for efflorescence on their worksheet. Groups then match each experimental sample explicitly to the anchoring phenomenon: 'Sample A = CAN fertiliser because...', 'Sample B = washing soda because...', 'Sample C = table salt because...'. A class discussion consolidates the three definitions. Learners add a 'Uses' column to their notes: deliquescent salts ($CaCl_2$) used as desiccants/drying agents in Kenyan tea processing factories in Kericho; efflorescent salts (washing soda) must be stored in sealed containers; stable salts (NaCl) ideal for food preservation.	reading, ask the class: 'Do you agree? What would you add or change?' WAIT TIME 10 seconds. Introduce the balanced equation for efflorescence: write $Na_2CO_3 \cdot 10H_2O \rightarrow Na_2CO_3 + 10H_2O$ on the board and ask: 'What type of change is this — physical or chemical? How do you know?' WAIT TIME 12 seconds, cold-call 3 learners. Then say: 'Now make the connection. Complete this sentence in your notebook: The CAN fertiliser clumped because..., the washing soda crusted because..., and the table salt was unchanged because...' Allow 4 minutes, then cold-call 3 learners to share complete sentences.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Genetics vocabulary Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	This is the most significant DQB update of the unit: the driving question receives its FIRST COMPLETE ANSWER. Each learner writes a full answer to the driving question on a green card (green = 'We can now answer this'): 'The CAN fertiliser granules clumped because CAN is a deliquescent salt — it absorbed water vapour from the humid air in the storeroom and dissolved in it. The washing soda crystals shrank and formed a white crust because $Na_2CO_3 \cdot 10H_2O$ is an efflorescent salt — it lost its water of crystallisation to the dry air, leaving behind anhydrous sodium carbonate powder. Table salt was unchanged because NaCl is neither hygroscopic nor efflorescent under normal conditions. To prepare and use salts safely in Kenya, we must store deliquescent salts (like CAN fertiliser) in sealed, airtight bags,	Distribute green cards. Say: 'Today is the day we have been building toward for six lessons. You now have enough evidence to write a full, scientific answer to our driving question. Write it on your green card — complete sentences, reference your evidence, name the three salts and their behaviours.' Allow 7 minutes. Collect all green cards. Read 3 aloud to the class, pausing after each to ask: 'Is this answer complete? What evidence supports it? Is anything missing?' Then direct learners to the full DQB: 'Look at every sticky note we have posted since Lesson 1. With your group, decide: Answered? Partially answered? Still investigating?' Allow 4 minutes of group discussion. Cold-call one spokesperson per group to report. Update the DQB live: move answered questions to the SOLVED column (use a green marker border), leave open	DQB Closure + Evidence Audit: Moving questions from 'investigating' to 'solved' gives learners a concrete, visual record of their growing scientific understanding. Writing the full driving-question answer requires learners to synthesise evidence from all six lessons — a high-order integration task. The public reading and critique of green-card answers models peer review and scientific argumentation (NGSS Practice 7).	Observable indicators: (1) Green card answer mentions all three salts by name and correctly labels each behaviour (deliquescent, efflorescent, unchanged). (2) Green card answer includes at least one piece of experimental evidence (e.g., mass change data or observation). (3) Green card answer includes a practical storage recommendation relevant to Kenya (CAN fertiliser, washing soda). Teacher uses green cards as an exit-ticket equivalent — any card missing two or more of these indicators identifies a learner who needs targeted support in Lesson 7's review opening.

	and efflorescent salts in sealed containers.' Learners also review ALL sticky notes posted on the DQB across Lessons 1–5, categorising them as 'Answered', 'Partially answered', or 'Still investigating'. The class updates the DQB together, moving answered questions to a 'SOLVED' column. Remaining open questions (preparation methods, specific uses) are flagged for Lessons 7–8.	questions in place. Say: 'Look at what is left. These are the questions that will drive Lessons 7 and 8. Write them in your notebook under the heading: Questions still to investigate.'		
Model Building Phase 📹 **VIDEO:** [Genetics vocabulary] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative particle-level model (begun in Lesson 1 and revised in each subsequent lesson). Today they add a new panel — a side-by-side diagram showing: (A) a deliquescent salt surface ($CaCl_2$) with water molecules from the air being attracted to and surrounding the ions at the surface, eventually forming a solution layer; (B) an efflorescent salt surface ($Na_2CO_3 \cdot 10H_2O$) with water molecules leaving the crystal lattice and escaping into the dry air, leaving a powdery residue; (C) a stable salt surface (NaCl) with water molecules in the air showing no significant interaction. Each panel must include: particle-level labelling (ions, water molecules, direction of movement using arrows), macroscopic observation linked to the particle event, and the name of the behaviour (deliquescent / efflorescent / stable). Learners peer-review one another's models using a 2-star-1-wish protocol before the lesson ends.	Say: 'Open your cumulative model. We are adding Panel 6 today — the most important panel yet, because it explains our phenomenon at the particle level.' Display a partial model on the board (ions arranged in a lattice, water molecules in the air shown as small clusters) and say: 'I am NOT going to complete this for you. Use your Frayer Model definitions and your experimental data to decide: which direction should the arrows point for each salt? What happens to the water molecules?' Allow 10 minutes for individual model drawing. After 8 minutes, say: 'Swap your model with the person next to you. Use 2-star-1-wish: find two things that are scientifically correct and one thing that could be improved or is missing.' Allow 3 minutes for peer review. Cold-call 3 learners to share one 'wish' (improvement suggestion) they received — ask the class: 'Do you agree with this suggestion? Why?' WAIT TIME 10 seconds. Close by returning to the anchoring phenomenon image on the DQB: 'Point to your model. Point to the photograph. Can your model now fully explain what we saw on Day 1?' Allow 30 seconds	Cumulative Particle Model Revision + 2-Star-1-Wish Peer Review: Adding a new evidence-based panel to a model that has grown across six lessons gives learners a tangible record of their conceptual development. The peer-review protocol (NGSS Practice 8 — Obtaining, Evaluating, and Communicating Information) requires learners to evaluate scientific accuracy, not just aesthetics, and to articulate what a correct model must include.	Observable indicators: (1) Model Panel 6 correctly shows arrows pointing INWARD (toward the salt) for deliquescent and OUTWARD (away from the salt) for efflorescent behaviour. (2) Water molecules are drawn as distinct particles, not as a continuous liquid. (3) Each panel is labelled with both the behaviour name and a macroscopic observation. (4) The peer-review 'wish' comment is scientifically substantive (not just aesthetic). Teacher photographs 4–5 models with a phone for use in the Lesson 7 opening review.

		of silent reflection, then cold-call 2 learners.		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PHENOMENON CONNECTION: Did every learner successfully connect all three experimental samples to the three parts of the anchoring phenomenon by name? Check the phenomenon-matching sentences in learner notebooks — if more than 30% of learners could not make all three connections independently, open Lesson 7 with a targeted re-examination of the class data table alongside the DQB photograph. (2) DQB QUALITY: Review the green cards collected at the end of Phase 4. Tally how many cards include all three required elements (correct behaviour labels, experimental evidence, storage recommendation). Use this count to decide whether Lesson 7 needs a 10-minute consolidation opener or can proceed directly to new content. (3) MODEL ACCURACY: Review the photographed models. Were arrows consistently drawn in the correct direction? Did learners distinguish between hygroscopic (gains mass but does not dissolve) and deliquescent (gains enough mass to dissolve) — or did they conflate the two? If conflation is widespread, add a brief clarification in Lesson 7. (4) SAFETY AND MANAGEMENT: Were all chemical samples disposed of correctly? Were goggles worn throughout the experiment? Note any safety issues for the laboratory register. (5) EQUITY CHECK: Were cold-call responses drawn from a diverse range of learners across gender and ability levels? Note any learners who did not contribute verbally across the lesson — ensure they are prioritised for cold-calling in Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners placed pre-weighed samples of anhydrous $CaCl_2$, $Na_2CO_3 \cdot 10H_2O$, and NaCl on watch glasses and left them exposed to classroom air for 20–25 minutes. They recorded mass changes at timed intervals and observed that $CaCl_2$ gained mass and became moist/sticky, $Na_2CO_3 \cdot 10H_2O$ lost mass and developed a white powdery surface, and NaCl showed no significant change.
What did I learn?	Salts behave differently when exposed to air: deliquescent salts (e.g., $CaCl_2$, CAN fertiliser) absorb water vapour from the air and can dissolve in it; efflorescent salts (e.g., $Na_2CO_3 \cdot 10H_2O$, washing soda) lose water of crystallisation to the air forming a white powder ($Na_2CO_3 \cdot 10H_2O \rightarrow Na_2CO_3 + 10H_2O$); stable salts (e.g., NaCl, table salt) show no significant interaction with atmospheric moisture under normal conditions.
How does this explain the phenomenon?	The CAN fertiliser granules clumped and became wet because CAN contains deliquescent salts that absorbed water vapour from the humid storeroom air and dissolved in it. The washing soda crystals shrank and formed a white crust because $Na_2CO_3 \cdot 10H_2O$ is efflorescent — it lost its water of crystallisation to the air. Table salt was unchanged because NaCl is stable in air. These properties determine how salts must be stored and used safely in Kenya: CAN fertiliser must be kept in sealed, airtight bags, and washing soda in sealed containers.

LESSON 7 (80 minutes): Applications of Salts in Daily Life in Kenya	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to outline and appreciate the applications of salts in day-to-day life in Kenya, connecting each application to the specific salt properties investigated in earlier lessons.
Knowledge	Learners will be able to: (i) identify specific salts used in agriculture, food industry, medicine, paper, paint, glass, and laundry industries; (ii) link the choice of a salt for a specific application to its type, solubility, and physical/chemical properties; (iii) connect the hygroscopic behaviour of CAN fertiliser and the efflorescence of washing soda — observed in the anchoring phenomenon — to their agricultural and industrial significance.
Skills	Learners will be able to: (i) search for, read, and synthesise information on the applications of salts from a structured text; (ii) present group findings clearly using a poster or summary chart; (iii) write concise explanatory statements linking salt properties to their uses; (iv) update the Driving Question Board with new evidence and career/societal connections.
Attitudes	Learners will appreciate the applications of salts in day-to-day activities and develop a sense of responsibility for safe and informed use of salts in Kenyan agriculture, food preservation, and medicine.
Key Inquiry Question	How do the properties of a salt determine the way it is used in Kenyan agriculture, food, medicine, and industry — and why is it important to choose the right salt for the right purpose?
Purpose in Storyline	In Lessons 1–6 learners established what salts are, how they are classified, how their solubility is determined, and how they behave in air (hygroscopic, deliquescent, and efflorescent). Lesson 7 is the culminating evidence-gathering lesson: learners now ask WHY each salt is chosen for its specific real-world purpose, directly connecting the properties they have investigated back to the three contrasting substances in the anchoring phenomenon — CAN fertiliser (deliquescent → must be stored sealed), washing soda (efflorescent → loses water and forms a crust), and table salt (neither → safe open storage). This lesson also populates the DQB with societal and career connections, preparing learners for the full explanation synthesis in Lesson 8.
Safety Notes	This lesson is a research-and-presentation activity — no hazardous chemicals are handled. If printed materials or digital devices are used, learners should handle school equipment responsibly (KICD Value: Responsibility). Remind learners that in real-world contexts, some salts used in industry (e.g., lead compounds in paint, concentrated fertiliser solutions) are hazardous and must be handled with appropriate PPE — this reinforces the PCI: Environmental Issues / Safety in the Laboratory.

B. LESSON OVERVIEW

Learners work in six expert groups, each assigned one application sector of salts in Kenya: (1) Agriculture — fertilisers in the Rift Valley, (2) Food Industry — fish preservation along Lake Victoria, (3) Medicine — oral rehydration salts and iron supplements, (4) Paper and Paint Industries, (5) Glass Industry, (6) Laundry Industry. Each group reads a structured sector text (provided on the ARES platform), identifies the specific salt(s) used, the property that makes each salt suitable, and how that property connects to findings from earlier lessons. Groups create a summary poster/chart and present to the class in a gallery walk. The lesson closes with a whole-class DQB update that maps every application back to the anchoring phenomenon and identifies career pathways connected to salts chemistry.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Conservation and the race to save biodiversity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown three images on the ARES platform: (A) a Rift Valley maize farmer pouring CAN fertiliser granules into a field; (B) a fishmonger at Kisumu's Dunga Beach rubbing salt onto fresh tilapia; (C) a nurse at a rural dispensary mixing oral rehydration salts (ORS) into water for a child. Learners individually write a prediction in their exercise books: 'For each image, identify the salt most likely being used, ONE property of that salt that makes it suitable for this purpose, and state whether that property is something you have already investigated in this unit.' Learners share predictions with a shoulder partner before the teacher takes cold-call responses.	Display the three images and READ the prediction prompt aloud. Say: 'You have spent six lessons gathering evidence about salts — their types, solubility, and behaviour in air. Before we read anything new today, I want you to reach back into that evidence and make a prediction. Write your ideas individually first — you have 4 minutes.' [WAIT — 4 minutes silent writing.] After 4 minutes: 'Share your prediction with the person next to you. Listen carefully — do you agree on the property? You have 2 minutes.' [WAIT — 2 minutes pair talk.] Cold-call 4 learners (aim for at least one from a rural/farming background): 'Achieng, what salt did you predict for the farmer and what property did you choose?' After each response, press: 'What evidence from our earlier lessons supports that prediction?' Do NOT confirm or correct answers yet — record all predicted properties on the board under the heading: 'Properties we THINK matter for each use.'	Strategy: Activation of Prior Knowledge through Prediction Mapping. By requiring learners to connect a specific property to a specific use BEFORE reading, this strategy surfaces existing mental models and creates a 'need to know' gap. Recording predictions publicly on the board gives learners an anchor to evaluate during the reading phase, making the eventual confirmation or revision of ideas personally meaningful.	Observable indicator: Each learner has written at least one salt name and one property in their book before pair discussion begins. Teacher scans books during the 4-minute window. Listen during cold-calls for whether learners reference evidence from earlier lessons (e.g., 'CAN is deliquescent, we saw that in the phenomenon') versus making unsupported guesses. Identify misconceptions (e.g., 'all salts preserve food') to target during the Explain phase.
Observe Phase 📹 **VIDEO:** Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12	Learners are organised into six Expert Groups (approximately 5 learners each). Each group is assigned one application sector and receives the corresponding structured reading text on the ARES platform (or as a printed handout). The six sectors are: (1) Agriculture — CAN, ammonium sulphate, and single super phosphate as fertilisers; role of nitrogen, phosphorus, and potassium; why CAN must be stored in sealed bags (links to deliquescence from the phenomenon). (2) Food Industry	Assign groups before the lesson and have materials ready on ARES. Introduce the task: 'Today you are expert researchers. Each group will become the class authority on one application sector of salts. Your job is to read carefully, complete the reading guide, and prepare a poster your classmates will learn from. You have 25 minutes for reading and poster preparation.' [WAIT — circulate continuously.] Visit each group at least twice. During visits, use targeted prompts: For Group 1 (Agriculture): 'You saw CAN clump	Strategy: Jigsaw Expert Reading with Structured Note-Taking. Dividing the class into sector experts means every learner reads deeply on one area and then teaches peers — a well-evidenced strategy for content retention. The four-column reading guide (Salt | Formula | Property | Phenomenon Connection) scaffolds the reading so learners are not simply collecting facts but are explicitly building explanatory links. The mandatory 'connection to phenomenon' column ensures the anchoring phenomenon remains	Observable indicator: Each group's completed reading guide has all four columns filled with accurate information for at least two salts. Teacher checks 'Connection to Phenomenon' column during circulation — if it is blank or vague (e.g., 'it is related'), prompt with: 'Be specific — which substance from the storeroom or laboratory bench does this remind you of, and why?' Poster quality check: the poster must name the key property (not just the use) — this distinguishes surface recall from conceptual understanding.

🔍 Search ARES for similar readings	— sodium chloride for preserving omena and tilapia at Lake Victoria; sodium nitrate/nitrite in processed meats; role of solubility and osmotic properties. (3) Medicine — oral rehydration salts (sodium chloride + potassium chloride + sodium citrate + glucose); iron(II) sulphate tablets for anaemia; zinc sulphate for child diarrhoea treatment; why ionic composition matters. (4) Paper and Paint Industries — sodium sulphate (salt cake) in paper pulping; barium sulphate as a white pigment in paint (links to insolubility rule from solubility lesson); titanium dioxide context. (5) Glass Industry — sodium carbonate (soda ash from Lake Magadi) as a flux to lower the melting point of silica; calcium carbonate to stabilise glass; sodium silicate. (6) Laundry Industry — sodium carbonate (washing soda) as a water softener; sodium hypochlorite as a bleach; links to efflorescence of washing soda observed in the phenomenon. Each group completes a structured reading guide with four columns: Salt Name | Chemical Formula | Key Property Used | Connection to Phenomenon or Earlier Lessons. After reading, the group prepares a summary poster (A3 paper or digital slide) with: sector title, two or three key salts, the property driving each use, and one sentence connecting the use to the phenomenon.	overnight in the phenomenon. Find the word in your text that describes this behaviour. Why does the farmer NEED to know this?' For Group 2 (Food): 'What property of sodium chloride makes it effective at preserving fish? Is that a physical or chemical property?' For Group 4 (Paint): 'Your solubility table from Lesson 4 — is barium sulphate soluble or insoluble? Why might that matter for a pigment in paint?' For Group 6 (Laundry): 'Your text mentions washing soda. Look back at the phenomenon description — what happened to the washing soda crystals overnight? Write that exact behaviour in your connection column.' If a group finishes early, prompt: 'Add a column — name ONE career in Kenya where a person would need to know this application.' After 25 minutes, call time: 'Posters up. We move into gallery walk in 2 minutes — leave one member at your poster to explain; the rest walk.'	the cognitive thread across all groups.	
Explain Phase 📹 **VIDEO:** Conservation and the race to save biodiversity Source: Khan	Gallery Walk (15 minutes): Five learners rotate clockwise through the five other groups' posters. One 'expert' remains at each poster. Rotating learners carry a sticky note and write ONE question and	For the gallery walk: 'Rotate when I say GO. You have 3 minutes at each poster. Read, ask the expert one question, and leave a sticky note with your question AND one connection to your own sector.'	Strategy: Gallery Walk with Sticky-Note Questioning, followed by Property→Use→Phenomenon Framework. The gallery walk distributes expertise across the class and creates peer	Observable indicator: During cold-calls, learners can name the specific property (not just the use) and connect it to the phenomenon without being prompted. Written synthesis statements should

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	ONE connection to their own group's sector on the note, which they leave on the poster. After the gallery walk, experts collect the sticky notes and share the most interesting question with the class. Whole-Class Synthesis (10 minutes): Teacher facilitates a structured class discussion using the 'Property → Use → Phenomenon' framework. For each of the three phenomenon substances (CAN, washing soda, table salt), the teacher asks learners to now provide a FULL explanation: What salt is it? What property does it have? How does that property determine both its atmospheric behaviour AND its practical use? Learners record a final synthesis statement in their books: 'The choice of a salt for a specific purpose depends on its [type / solubility / behaviour in air] because...' — completed individually.	[Timer visible. Rotate every 3 minutes. Circulate and listen for misconceptions.] After gallery walk, call the class back: 'Experts — read out the most surprising question left on your poster.' Take 3–4 questions, write them on the board. Then say: 'Let us now use ALL the evidence from today — and from our six lessons before today — to answer the three parts of our driving question about the phenomenon.' Display the phenomenon description on the screen. Cold-call 6 learners (one per group): 'Mwangi — your group studied agriculture. CAN clumped overnight. In ONE sentence, connect the property of CAN to why a Rift Valley farmer must store it in sealed bags.' After each response, ask the class: 'Does anyone want to add or challenge?' [WAIT 10 seconds after each response.] After all three phenomenon substances are explained, say: 'Now write your synthesis statement. You have 3 minutes — no talking.' [WAIT — 3 minutes silent writing.] Cold-call 3 learners to read their statements. Provide explicit verbal feedback: 'Excellent — Njeri has named the property AND connected it to a real consequence for the farmer. That is the level of explanation we are building towards in Lesson 8.'	accountability (learners must be able to explain their poster to visitors). The structured synthesis framework ('Property → Use → Phenomenon') prevents learners from treating applications as isolated facts by requiring them to trace every use back to a property and every property back to observable evidence — the core of scientific sensemaking. The individual synthesis statement consolidates learning in writing, supporting literacy across the curriculum.	contain: a named salt, a named property, and a stated consequence — teacher collects 6–8 books at random at the end of the lesson to check for depth and accuracy. Sticky notes on posters reveal the level of cross-sector thinking — if most notes only restate information rather than ask questions, teacher plans a bridging activity for Lesson 8.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Conservation and the race to save biodiversity Source: Khan Academy (English	Each expert group receives three sticky notes in different colours: (GREEN) 'We can now explain this about the phenomenon'; (YELLOW) 'We now understand this property and its use'; (PINK) 'New question or career connection raised by today's lesson.' Groups write their notes and a class representative places	Hand out the three sticky-note colours to group leaders. Say: 'You have 4 minutes as a group — fill in all three colours. Be specific: name the salt, name the property, name the application.' [WAIT — 4 minutes.] Call one representative per group to place notes on the DQB. After all notes are placed, step back and say: 'Look at our	Strategy: Colour-Coded DQB Update for Evidence Tracking. Using three distinct colours maps directly onto the three epistemic states of a scientific investigation: what is now explained (green), what properties are understood (yellow), and what remains open or extends to new contexts (pink). This visual differentiation helps	Observable indicator: Green notes make specific, accurate claims (e.g., 'CAN is deliquescent — it absorbs water vapour from air, so granules clump. Farmers must seal bags'). Vague green notes (e.g., 'we understand salts now') indicate surface-level understanding — teacher flags these groups for targeted follow-up

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	them on the DQB under three columns: 'Evidence Gathered', 'Properties Explained', and 'New Questions / Career Links'. The class briefly reviews the full DQB and identifies which parts of the original driving question can now be substantially answered and which parts still require the final lesson.	board. In silence, read through the green notes — the things we can now explain.' [WAIT 15 seconds.] Then ask: 'Raise your hand if you think we can now explain WHY the CAN fertiliser clumped.' [Count hands.] 'Raise your hand if you can explain the washing soda crust.' [Count hands.] 'Raise your hand if you can explain why table salt was unchanged.' [Count hands.] Point to the pink notes: 'These new questions and career connections are exactly what makes chemistry more than a school subject — they connect what we learn here to what Kenyans do every day. We will use these in Lesson 8 to build our final complete explanation.' Take a photograph of the updated DQB for the ARES platform record.	learners develop metacognitive awareness of their own understanding — a key Competency-Based Education skill — and surfaces the remaining gaps that Lesson 8 must address.	in Lesson 8. Pink notes with career connections (e.g., 'Agricultural extension officer in Nakuru needs to know which fertiliser to recommend based on storage conditions') indicate higher-order transfer thinking.
Model Building Phase 📹 **VIDEO:** Conservation and the race to save biodiversity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually revise their cumulative model of the phenomenon in their exercise books. The model should now include ALL of the following layers built across Lessons 1–7: (i) Identification of each substance as a salt (type); (ii) Solubility classification; (iii) Behaviour in air (hygroscopic / deliquescent / efflorescent / neutral); (iv) At least ONE named real-world application connected to each substance's properties; (v) An arrow or annotation showing the link between property and application. Learners also add a 'Why it matters in Kenya' annotation — one sentence per substance explaining a real Kenyan consequence of the property (e.g., 'Rift Valley farmers lose money if CAN is stored in open bags because it absorbs moisture and the granules stick together, making	Say: 'Open your cumulative model from previous lessons. Today you are adding the final layer before our big explanation in Lesson 8 — the application layer. Your model must now show not just WHAT each substance does in air, but WHY that matters to a real Kenyan — a farmer, a fishmonger, a nurse, or an industry worker. You have 6 minutes.' [WAIT — 6 minutes. Circulate and look for the five required layers.] After 6 minutes, cold-call 3 learners to hold up their model and describe one new annotation they added today. Ask the class: 'Thumbs up if your model now connects a property to a real Kenyan use. Thumbs sideways if you are not sure how to make that connection.' For thumbs-sideways learners: 'Stay for 2 minutes after the lesson and we will look at your model together.' Close the lesson: 'Next	Strategy: Cumulative Model Revision with Contextual Annotation. Adding the 'Why it matters in Kenya' annotation layer transforms the model from a purely descriptive scientific diagram into an applied explanatory tool — directly reflecting the KICD competency of appreciating the applications of salts in day-to-day life (SLO e). Requiring learners to name a specific Kenyan actor (farmer, fishmonger, nurse) grounds abstract chemistry in human consequence, reinforcing Communication and Collaboration competencies and societal values.	Observable indicator: The revised model contains all five required layers. Teacher specifically checks that the 'application' layer names a specific salt and a specific property (not just 'it is useful'). The 'Why it matters in Kenya' annotation must name a context, a person/industry, and a consequence — this three-part structure is the benchmark for readiness for Lesson 8's full synthesis task. Models lacking the property-to-application arrow are flagged for the opening review of Lesson 8.

	spreading difficult.').	lesson — Lesson 8 — you will use everything in this model, everything on the DQB, and everything from our six experiments to write your final complete explanation of the phenomenon. Come ready.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the gallery walk, did learners' sticky-note questions demonstrate cross-sector thinking — connecting, for example, the solubility of barium sulphate to its use in paint — or did most questions remain within their own sector? If cross-sector thinking was weak, plan a brief 'property matching' warm-up at the start of Lesson 8. (2) Were the 'connection to phenomenon' columns in the reading guides completed with specificity? If learners in the Laundry Group (washing soda) or Agriculture Group (CAN) were unable to link their reading back to the phenomenon without significant prompting, this signals that the earlier lessons on hygroscopic/deliquescent/efflorescent behaviour (likely Lesson 5 or 6) may need reinforcement before the Lesson 8 synthesis. (3) Review the 6–8 collected exercise books: do written synthesis statements contain a named salt, a named property, AND a stated consequence? If fewer than 70% of sampled books meet this three-part benchmark, the opening of Lesson 8 must include a structured whole-class modelling exercise before learners attempt their final explanation. (4) Count the pink DQB sticky notes with career connections — these indicate learners who are beginning to see chemistry as personally relevant. Share two or three of the most compelling career connections at the start of Lesson 8 to reinforce motivation and the societal value of the unit. (5) Identify any learners who gave thumbs-sideways during the model-building phase and ensure targeted support before or at the start of Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners read structured texts and examined images showing: CAN fertiliser being applied on Rift Valley maize farms and stored in sealed bags; sodium chloride being rubbed onto tilapia and omena at Kisumu's Dunga Beach; ORS being prepared at a rural dispensary; soda ash from Lake Magadi being used in glass manufacture; washing soda being added to laundry water; and barium sulphate being used as a pigment in paint.
What did I learn?	The choice of a salt for a specific application depends on its type, solubility, and behaviour in air. Deliquescent salts like CAN (calcium ammonium nitrate) must be stored in sealed containers because they absorb water vapour — the same property that caused granules to clump in the phenomenon. Efflorescent salts like washing soda ($Na_2CO_3 \cdot 10H_2O$) lose water of crystallisation in dry air, forming a white powdery crust — this is why it is effective as a laundry water softener but must be stored in airtight containers. Insoluble salts like barium sulphate are valuable as paint pigments precisely because they do not dissolve. Soluble salts like sodium chloride and ORS components are effective in food preservation and medicine because they readily dissolve and interact with biological systems.
How does this explain the phenomenon?	All three substances in the phenomenon are salts, but their different behaviours in air — and their different real-world uses — are both explained by the same underlying properties. CAN is deliquescent: it absorbs water vapour from the air, which is why granules clump overnight AND why fertiliser manufacturers and Rift Valley farmers must store it in sealed bags to prevent caking and nutrient loss. Washing soda is efflorescent: it loses water of crystallisation to the dry air, forming anhydrous sodium carbonate as a white crust AND this same water-releasing property makes it a powerful laundry water softener by reacting with calcium and magnesium ions. Table salt is neither hygroscopic nor efflorescent under normal conditions: it does not react with air moisture, which is why it remains unchanged in the kitchen store AND why it is a reliable, stable preservative for fish and food. The property that determines how a salt behaves in air is the same property that determines how — and whether — it can be safely and

	effectively used in Kenyan agriculture, food, medicine, and industry.

LESSON 8 (80 minutes): Unit Synthesis, Model Revision and Summative Assessment	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate and synthesise all learning from the Introduction to Salts unit by revising cumulative explanatory models, completing the Driving Question Board, and demonstrating understanding through a summative assessment task.
Knowledge	Learners will be able to: (a) classify different salts based on their properties; (b) describe the behaviour of salts when exposed to air; (c) outline methods of preparation appropriate to solubility; (d) outline applications of salts in day-to-day life in Kenya.
Skills	Learners will be able to: (b) prepare salts using appropriate methods in the laboratory; and synthesise written balanced chemical equations, annotated diagrams, and evidence-based explanations into a coherent final explanatory model.
Attitudes	Learners will: (e) appreciate the applications of salts in day-to-day life, and demonstrate responsibility by accurately representing scientific evidence in their final models and assessment responses.
Key Inquiry Question	1. How do salts behave when exposed to air? 2. How can salts be prepared in the laboratory?
Purpose in Storyline	This is the final lesson of the unit. Learners have gathered evidence across seven lessons — from classifying salt types, testing solubility, preparing salts by multiple methods, and investigating hygroscopic, deliquescent and efflorescent behaviour. In this lesson they bring all threads together: they revise the cumulative model they began in Lesson 1, close out the Driving Question Board, and demonstrate their full understanding by completing a structured summative task that returns them to the anchoring phenomenon — the CAN fertiliser bag, washing soda crystals, and unchanged table salt — and asks them to explain, prepare, predict and apply.
Safety Notes	No new chemicals are used in this lesson. If learners handle any salt samples during model revision, standard laboratory safety applies: no tasting, wash hands after handling. Dispose of any salt residues into the designated waste container. Remind learners that the responsible handling of fertilisers and washing soda in real Kenyan contexts (farms, homes, laundries) reflects the same safety principles practised throughout the unit.

B. LESSON OVERVIEW
Lesson 8 is the capstone lesson of the Introduction to Salts unit. It opens with a structured 'Return to Phenomenon' provocation in which learners view the original anchoring image — the open CAN fertiliser bag, the crusted washing soda container, and the unchanged table salt — and are asked silently to draft a full scientific explanation using everything they now know. Groups then compare their Lesson 1 prediction models with their current cumulative models in a gallery walk, annotating growth in understanding. The full class revisits the Driving Question Board: answered questions are confirmed with evidence citations, genuinely unresolved questions are honoured and flagged for future inquiry. The lesson closes with a structured summative assessment task in which individual learners identify an unknown salt, propose its preparation pathway with a balanced equation, predict its behaviour in air and describe one Kenyan industrial or agricultural use. Teacher facilitation throughout uses Socratic questioning, strategic cold-calling and explicit wait time to ensure every learner is drawn into the final sensemaking conversation.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners are shown the original	Display the phenomenon image	Strategy: 'Then vs. Now' Pre-Write	Observable indicator: Each

📹 **VIDEO:** [Meaningfully composing functions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	anchoring phenomenon image on the board or screen: the open CAN fertiliser bag with clumped, sticky granules; the washing soda container with its white powdery crust and shrunken crystals; and the unchanged table salt from the kitchen store. Each learner receives their Lesson 1 prediction card (returned by teacher). Working silently and individually for 5 minutes, they write a full scientific explanation on a new card — 'What I can now explain' — covering: (1) why CAN clumped (deliquescence / hygroscopic behaviour, ionic nature, solubility); (2) why washing soda developed a white crust and shrank (efflorescence, loss of water of crystallisation, $Na_2CO_3 \cdot 10H_2O \rightarrow Na_2CO_3 \cdot H_2O + 9H_2O$); (3) why table salt was unchanged (NaCl is neither hygroscopic, deliquescent nor efflorescent under normal Kenyan ambient humidity conditions). Learners then turn and compare their new card with their Lesson 1 card, underlining every concept they could NOT have written in Lesson 1.	without any labels. Say exactly: 'You saw this on Day 1. You had questions. You gathered evidence. Now — without talking, without notes — write the full scientific story of what happened to each of these three salts overnight. You have 5 minutes.' Set a visible timer. Circulate silently; do not answer questions — redirect with: 'Check your cumulative model.' After 5 minutes, say: 'Underline every idea on your new card that you could NOT have written on Day 1. Those underlines are your learning.' Cold-call 3 learners to read one underlined phrase each. Record the three phrases on the board under the heading: 'Evidence of growth.' Apply 12 seconds of WAIT TIME after the prompt before calling on anyone.	Comparison. By physically holding their Lesson 1 prediction card alongside their new explanation card, learners make their own conceptual growth visible and tangible. This metacognitive strategy builds self-efficacy and consolidates declarative knowledge by requiring learners to articulate it without scaffolding — mirroring the NGSS practice of Obtaining, Evaluating and Communicating Information.	learner's new explanation card must contain at minimum — (a) the correct technical term for CAN's behaviour (deliquescent or hygroscopic); (b) the word 'efflorescence' or 'water of crystallisation' for washing soda; (c) a statement explaining why NaCl is unaffected. Teacher scans cards during circulation. Any learner whose card is missing one or more of these is flagged for targeted support during the gallery walk phase.
Observe Phase 📹 **VIDEO:** [Meaningfully composing functions](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)](#) Source: CK-12	Groups of four display two posters side by side on the wall or desk: their Lesson 1 initial model (a rough sketch of the phenomenon with guesses) and their final cumulative model built incrementally across Lessons 1–7. The final model must show, in one annotated diagram: (a) salt classification tree (normal, acidic, basic, double) with one Kenyan example per type; (b) solubility rules summary (all nitrates soluble; sulphates — exceptions include barium, lead, calcium sulphate; chlorides — exceptions include	Before rotation begins, say: 'Your job at each poster is to be a scientific peer reviewer — not a judge. Write what EVIDENCE you see in the model, and one genuine question the model raises for you.' Set timer for 6 minutes per rotation (two rotations = 12 minutes total). During rotations, focus your own movement on groups whose Lesson 1 model was weakest — ask: 'Point to where your model shows the difference between deliquescence and hygroscopic behaviour. What evidence from our experiments made you draw it	Strategy: Comparative Model Gallery Walk. Placing the Lesson 1 model physically beside the final model externalises cognitive growth and allows peers to act as critical readers — an authentic scientific practice. The sticky-note protocol (evidence vs. question) separates observation from inference, reinforcing the NGSS practice of Developing and Using Models. The Kenyan applications panel ensures the model is grounded in real, local contexts rather than abstract chemistry.	Observable indicator: Final cumulative model must contain all five components listed in the learner experience. Teacher uses a rapid 5-point checklist on a clipboard during the gallery walk: classification tree (1), solubility rules (2), preparation pathways with equations (3), atmospheric behaviour triangle with phenomenon substances correctly placed (4), Kenyan applications panel (5). Any group missing two or more components is invited to a brief 2-minute targeted conference at their poster before the DQB

🔍 Search ARES for similar readings	lead and silver chloride; carbonates — insoluble except ammonium and Group 1); (c) preparation pathway flowchart (direct synthesis, acid + metal, acid + base, acid + carbonate/hydrogen carbonate, precipitation) with one balanced equation per pathway; (d) atmospheric behaviour triangle (hygroscopic / deliquescent / efflorescent) with the three phenomenon substances placed correctly; (e) Kenyan applications panel (agriculture — CAN in Rift Valley maize farms; food — NaCl preservation of nyama choma and drying fish at Lake Victoria; medicine — saline solutions at Kenyatta National Hospital; laundry — washing soda in Nairobi households). Groups rotate to two other groups' posters, leaving one sticky note of 'Evidence I see' and one sticky note of 'A question I still have.'	that way?' After rotations, facilitate a 3-minute whole-class debrief: cold-call one spokesperson per group (3 groups × 1 person) to share the most surprising 'Evidence I see' sticky note they received. Apply 10 seconds of WAIT TIME between each share.		phase begins.
Explain Phase 📹 **VIDEO:** Meaningfully composing functions Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Hydrolysis of Salts: Equations (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The full Driving Question Board — maintained on the classroom wall since Lesson 1 — is brought into focus. All original question cards (posted by learners in earlier lessons) are visible. The teacher facilitates a structured whole-class review in three rounds: Round 1 — 'Answered with evidence': Learners identify questions they can now answer fully. A volunteer reads the question aloud, then a different learner provides the evidence-based answer and names which lesson or experiment generated the evidence. The card is moved to the 'Explained' column. Round 2 — 'Partially answered': Questions where learners have partial understanding are moved to a 'Still	Read the driving question aloud slowly and completely. Say: 'Seven lessons ago, this question puzzled you. Let's see what we know now.' Hand the DQB marker to a learner and say: 'You are the curator. Read each card — class, thumbs up if we can answer it fully with evidence.' For each 'thumbs up' question, say: 'Who can give the answer AND name the lesson or experiment that gave us the evidence?' Apply 12 seconds of WAIT TIME. Cold-call one learner for the answer, then cold-call a SECOND learner to name the evidence source — do not accept 'I don't know' without a 'Phone a Friend' prompt to a peer. For the consensus answer to the driving question, say: 'Give me exactly	Strategy: DQB Triaged Closure with Consensus Scribing. The three-round DQB review (answered / partial / new curiosities) models how scientific understanding grows iteratively — some questions are closed, some invite deeper inquiry. The 'Future Chemistry' column affirms that unresolved questions are intellectually valuable, not failures. Consensus scribing externalises the class's collective explanatory model and provides a visible anchor for the summative task — mirroring the NGSS practice of Constructing Explanations.	Observable indicator: The class consensus answer written on the board must, at minimum, use the terms 'deliquescent' or 'hygroscopic' for CAN, 'efflorescence' and 'water of crystallisation' for washing soda, and include a statement about NaCl's stability. If the spontaneous class answers are missing any of these, the teacher inserts a targeted prompt: 'We used a specific chemistry term for this in Lesson 5 — what was it?' Do not supply the term; wait for it to emerge from the class.

	building' column with a brief note on what further evidence would be needed. Round 3 — 'New curiosities honoured': Any genuine new questions that arose during the unit (e.g. 'Why does CAN fertiliser sometimes cause soil acidification over many seasons?') are written on new cards and placed in a 'Future Chemistry' column, signalling that science is an ongoing process. The driving question itself — 'Why did the fertiliser granules get wet and clump together overnight...' — is read aloud by the teacher in full. The class votes by show of hands: 'Are we ready to answer this question fully and confidently?' The teacher scribes a class consensus answer on the board in three sentences.	three sentences — one for CAN, one for washing soda, one for table salt. Go.' Scribe the three sentences verbatim as given by the class, editing only for scientific accuracy, and leave them on the board for the duration of the summative task.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Meaningfully composing functions] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Each learner independently completes a structured summative assessment task on a printed or digital task sheet. The task is framed around a new but parallel Kenyan scenario to ensure transfer — not simple recall: 'A Kenyan agrochemical company in Nakuru is developing a new fertiliser product. Laboratory analysis shows the fertiliser contains the salt X, which forms when sulphuric acid reacts with ammonia solution. (a) Name salt X and classify it as normal, acidic, basic or double. (b) Write a balanced chemical equation for its preparation. (c) The company stores salt X in open warehouses near Lake Nakuru where humidity is high. Predict and explain its likely behaviour in air, using correct terminology. (d) Describe ONE other industrial or agricultural use of a salt covered in this unit	Distribute task sheets. Say: 'This is your individual opportunity to show what YOU understand — not your group, not your model poster alone. You may look at your final model poster. You may not look at your textbook or exercise book. You have 20 minutes. Begin.' Set a visible timer. Circulate silently for the first 10 minutes — do not answer content questions; redirect with: 'What does your model show about that?' At the 10-minute mark, give a quiet time-check: 'Ten minutes remaining — check that you have attempted all four parts.' At 5 minutes remaining: 'Five minutes — begin checking your equations are balanced.' At time: 'Pens down. Read your answer to part (c) silently one more time and make one final edit if needed — 60 seconds.' Collect all task sheets.	Strategy: Transfer Task with Partial Scaffold. The summative task is deliberately set in a new Kenyan context (Nakuru, ammonia + sulphuric acid → ammonium sulphate) that was not the primary focus of any single lesson, requiring learners to transfer and apply — not simply recall. Allowing reference to their own cumulative model (but not textbooks) rewards learners who engaged consistently with model-building across the unit and assesses authentic scientific reasoning. This aligns with the NGSS practice of Constructing Explanations and Designing Solutions.	Observable indicator for teacher marking: (a) Salt X correctly identified as ammonium sulphate, $(NH_4)_2SO_4$, classified as a normal salt — 1 mark. (b) Balanced equation: $H_2SO_4 + 2NH_3 \rightarrow (NH_4)_2SO_4$ — 2 marks (1 for correct products, 1 for balance). (c) Prediction: ammonium sulphate is hygroscopic; in high-humidity conditions near Lake Nakuru it will absorb moisture from the air, causing granules to become damp and clump — correct use of 'hygroscopic' required for full marks — 2 marks. (d) Any valid salt with correct preparation method and Kenyan use — e.g. NaCl prepared by evaporation of seawater at Mombasa coast / used for food preservation; or calcium ammonium nitrate prepared by neutralisation of nitric acid with calcium hydroxide / used as fertiliser on maize farms in the Rift

	that is relevant to Kenya, naming the salt, its preparation method and its use.' Learners work silently and individually for 20 minutes. They may refer to their final cumulative model poster only — not to textbooks or notes — to simulate authentic scientific reasoning from internalised understanding.			Valley — 2 marks. Total: 7 marks.
Model Building Phase 📹 **VIDEO:** [Meaningfully composing functions] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Hydrolysis of Salts: Equations (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	After task sheets are collected, learners participate in a structured 'Model Evolution Share' — a brief, celebratory closure ritual. Each group selects ONE annotation from their final cumulative model that they are most proud of — the one piece of science that best explains the anchoring phenomenon — and a spokesperson reads it aloud to the class in one sentence. The teacher then displays (or reads aloud) the full driving question one final time, and the class reads the three-sentence consensus answer from the board in unison. Learners write a personal 'Learning Legacy' sentence on a sticky note — 'One thing I will now notice differently in the world because of this unit' — and post it on a designated wall section. Examples might include: 'I will now check whether fertiliser bags on our shamba in Kericho are properly sealed'; 'I will understand why washing soda in our home gets crusty'; 'I will be able to explain to my father why the CAN granules clumped.' The teacher closes by naming the KICD core competencies the class demonstrated across the unit and affirming the values of responsibility and scientific curiosity.	After collecting task sheets, say: 'Before we finish — your models told a scientific story across eight lessons. I want to hear the line you are most proud of.' Give groups 90 seconds to select their sentence. Cold-call all groups in sequence — no WAIT TIME here; pace is upbeat and celebratory. Then say: 'Let's read our answer to the driving question together — the question that started everything.' Point to the board and lead the unison reading. For the sticky note, say: 'One sentence. What will you notice differently — at home, on the farm, in the kitchen — because of what you now understand about salts? Write it. Mean it. Post it.' As learners post notes, read 3–4 aloud. Close with: 'You started this unit asking why a bag of fertiliser got wet overnight. You now have the chemistry to answer that — and to apply it to salts you have never seen before. That is what it means to think like a chemist.'	Strategy: 'Learning Legacy' Sticky Note and Unison Consensus Reading. The 'Learning Legacy' prompt requires learners to connect unit chemistry to their own lived Kenyan contexts — sustaining and extending the anchoring phenomenon into personal meaning. Unison reading of the consensus answer creates a shared moment of intellectual closure and reinforces the language of chemistry through repeated oral production. Together these strategies fulfil the NGSS practice of Obtaining, Evaluating and Communicating Information, and honour KICD's value of responsibility by anchoring chemistry in real-world accountability.	Observable indicator: Each sticky note must contain a plausible real-world Kenyan context connected to the chemistry of salts. Teacher reads all posted sticky notes after class — any note that is purely abstract ('I learned about salts') is noted for follow-up in the next sub-strand introduction, where the teacher can invite that learner to make the connection more concrete. A minimum of 80% of sticky notes should name a specific context (a place, a food, a farm practice, a household item) — this is the teacher's informal retention and transfer indicator for the unit.

D. TEACHER REFLECTION
After Lesson 8, reflect on the following questions before closing the unit record: (1) MODEL GROWTH — When comparing Lesson 1 models to final models, were all five components present (classification, solubility, preparation, atmospheric behaviour, applications)? Which component was most consistently underdeveloped, and what would you strengthen in the instructional sequence for next year? (2) DQB CLOSURE — Were there questions on the DQB that the class could not answer confidently, even in Lesson 8? If so, which lesson in the sequence should be extended or redesigned to address that gap? (3) SUMMATIVE TASK — What percentage of learners correctly identified ammonium sulphate as a normal salt and wrote a balanced equation? If fewer than 70% succeeded on the equation, consider adding a dedicated equation-balancing practice session in Lessons 5 or 6 next cycle. (4) PHENOMENON CONNECTION — Did learners' 'Learning Legacy' sticky notes demonstrate genuine transfer to Kenyan agricultural, domestic and industrial contexts, or did responses remain abstract? If abstract, consider introducing more frequent 'connect to your world' prompts in earlier lessons. (5) EQUITY — Were there learners who contributed minimally to group model-building but performed differently (higher or lower) on the individual summative task? Use this discrepancy to inform grouping strategies and formative checkpoints in the next unit. (6) KICD COMPETENCIES — Note specific moments in this lesson where Communication and Collaboration and Digital Literacy were evidenced. Record one concrete example of each for your professional portfolio.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed (in Lesson 1) that CAN fertiliser granules left overnight in the school storeroom had clumped together and felt wet and sticky without any water being added; that washing soda crystals on the laboratory bench had developed a white powdery crust and appeared to have shrunk; and that ordinary table salt ($NaCl$) in the kitchen store was completely unchanged. All three substances are salts, yet their responses to the same air were strikingly different.
What did I learn?	Learners have learned that: (1) Salts are formed when hydrogen atoms in an acid are fully or partially replaced by a positive ion, and are classified as normal, acidic, basic or double salts. (2) Solubility follows predictable rules — all nitrates are soluble; all sulphates are soluble except barium sulphate, lead sulphate and calcium sulphate (slightly soluble); all chlorides are soluble except lead chloride and silver chloride; all carbonates are insoluble except ammonium carbonate and Group 1 carbonates. (3) The preparation method chosen depends on the solubility of the salt: soluble salts via direct synthesis, acid-metal, acid-base or acid-carbonate/hydrogen carbonate reactions; insoluble salts via precipitation. (4) Salts behave differently in air: hygroscopic salts absorb moisture without dissolving; deliquescent salts absorb enough moisture to dissolve in it (CAN/calcium ammonium nitrate); efflorescent salts lose water of crystallisation to the air forming a white powdery anhydrous or partially hydrated crust (washing soda, $Na_2CO_3 \cdot 10H_2O \rightarrow Na_2CO_3 \cdot H_2O + 9H_2O$); some salts such as $NaCl$ are unaffected under normal conditions. (5) The choice of salt for a specific use — in Kenyan agriculture, food preservation, medicine, laundry and industry — depends on the type and properties of the salt.
How does this explain the phenomenon?	Learners can now fully explain the anchoring phenomenon: CAN fertiliser granules clumped and became wet because calcium ammonium nitrate is deliquescent (or strongly hygroscopic) — its ionic components attract water molecules from the humid storeroom air, and the absorbed water causes granules to stick together, reducing fertiliser effectiveness if not stored in sealed containers. Washing soda crystals shrank and formed a white crust because $Na_2CO_3 \cdot 10H_2O$ is efflorescent — it spontaneously loses nine of its ten water molecules of crystallisation to the drier laboratory air, producing the anhydrous or monohydrate form as a white powder on the surface. Table salt ($NaCl$) remained unchanged because it is neither hygroscopic, deliquescent nor efflorescent under typical Kenyan ambient humidity; its crystal lattice does not attract or release water under normal conditions. Understanding these properties — and the preparation pathways and solubility rules that underpin them — explains why farmers in the Rift Valley must seal CAN bags, why washing soda tubs must be kept closed, and why $NaCl$ is the reliable everyday preservative used across Kenya from Mombasa fish markets to upcountry kitchens.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.